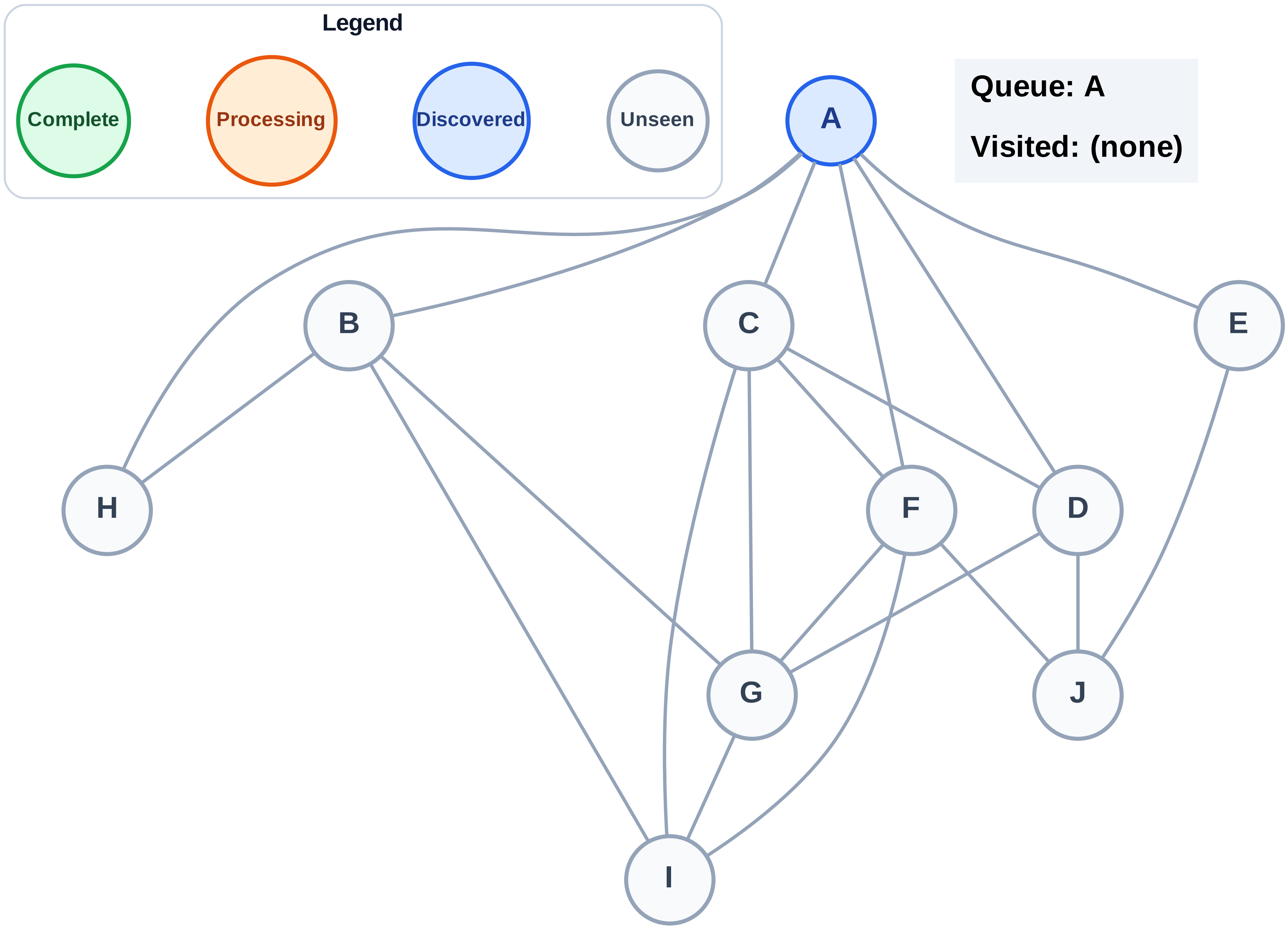
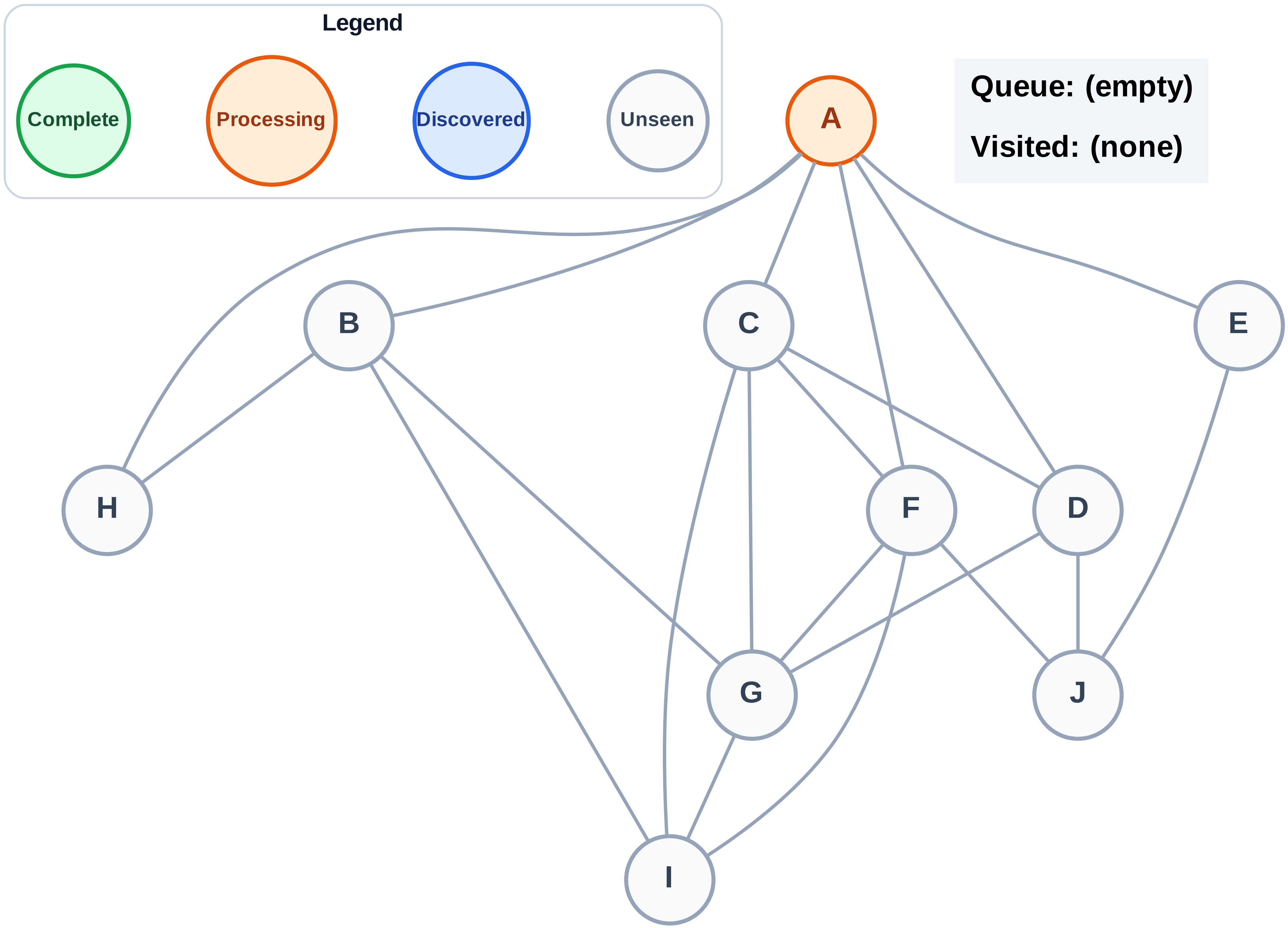
BFS Traversal

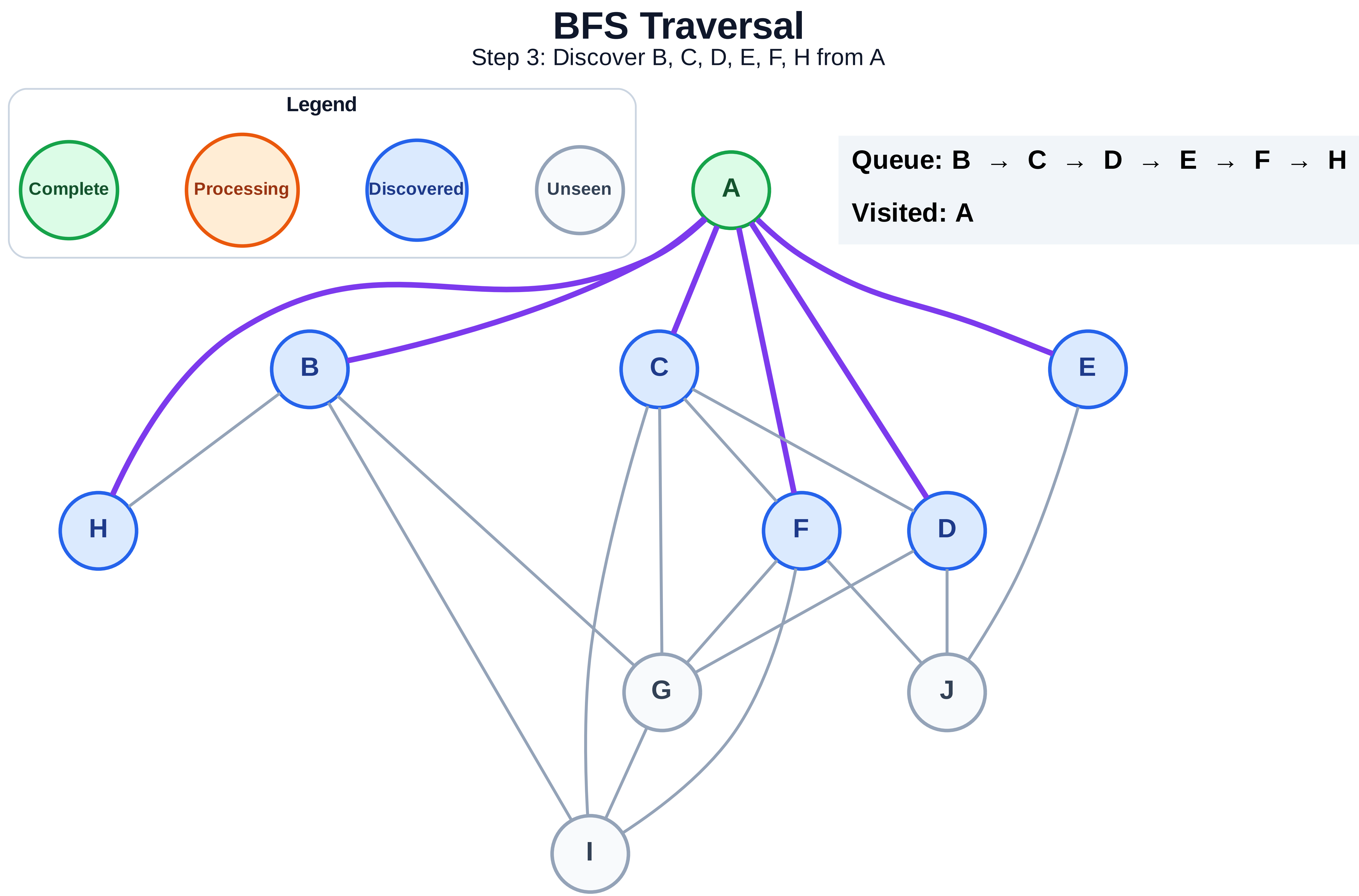
Step 1: Start at A



BFS Traversal

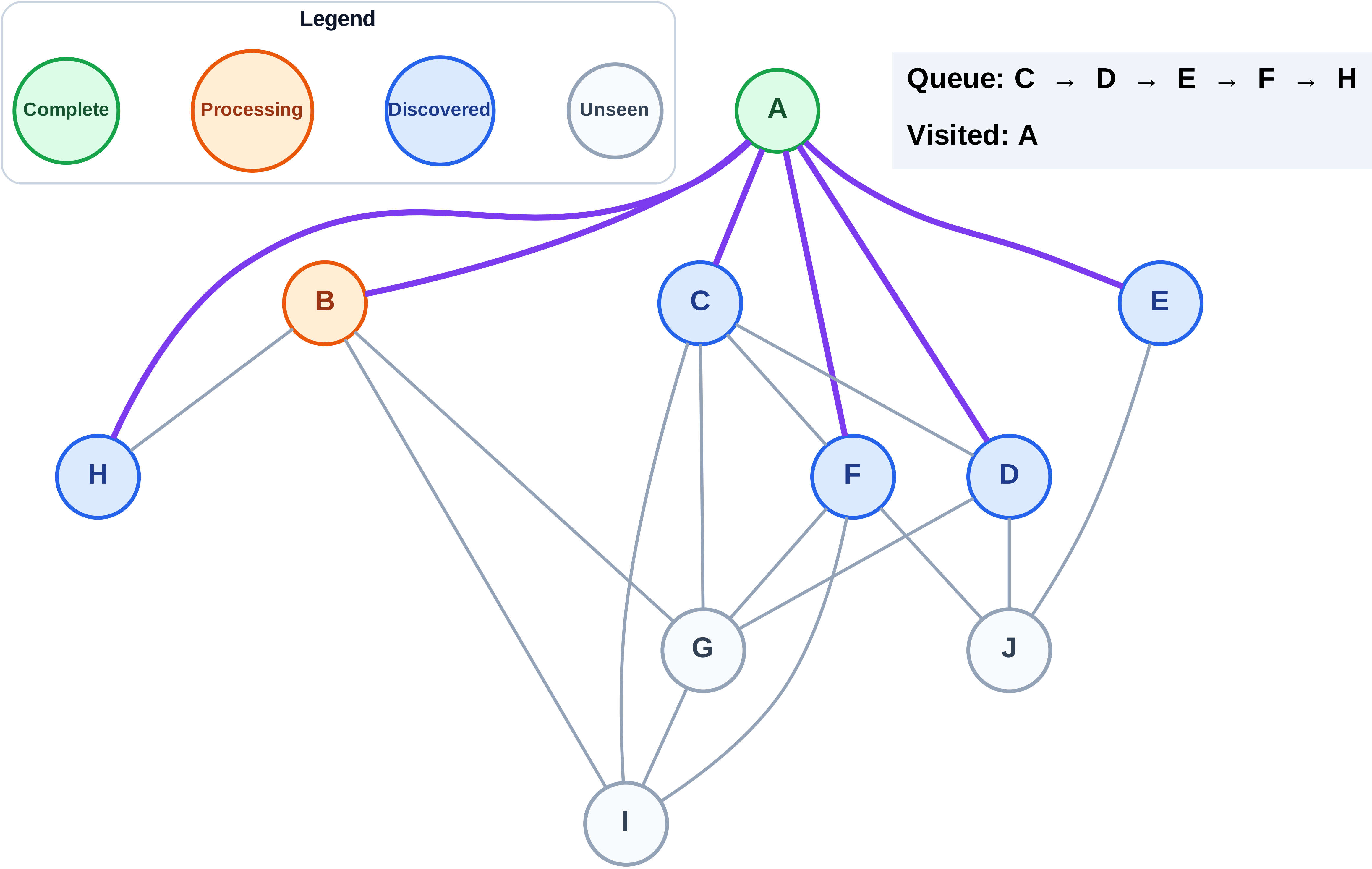
Step 2: Process node A




BFS Traversal
Step 3: Discover B, C, D, E, F, H from A
Legend
Complete
Processing
Discovered
Unseen
Queue: B → C → D → E → F → H
Visited: A
A
B
C
E
H
F
D
G
J
I

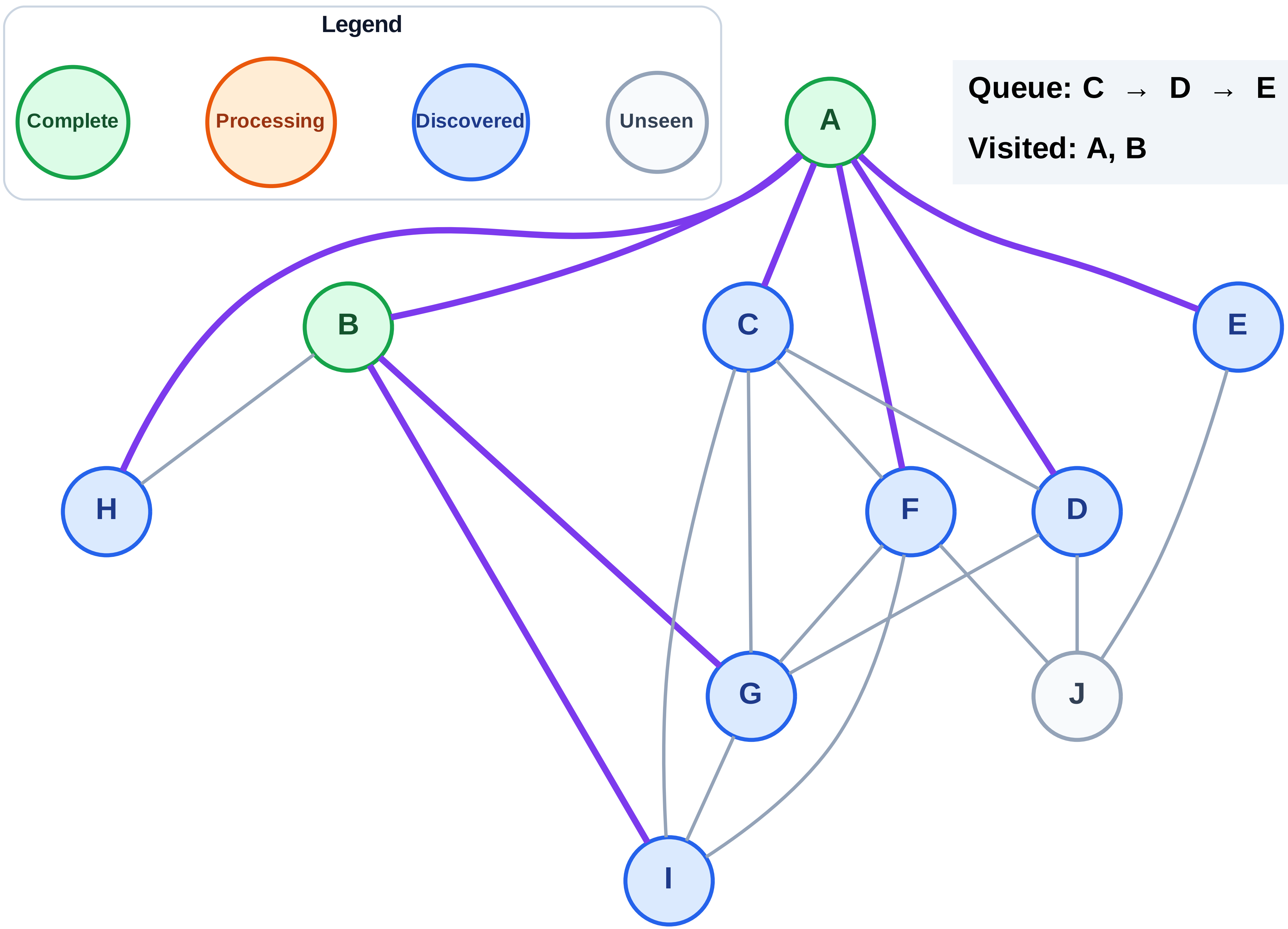
BFS Traversal

Step 4: Process node B



BFS Traversal

Step 5: Discover G, I from B

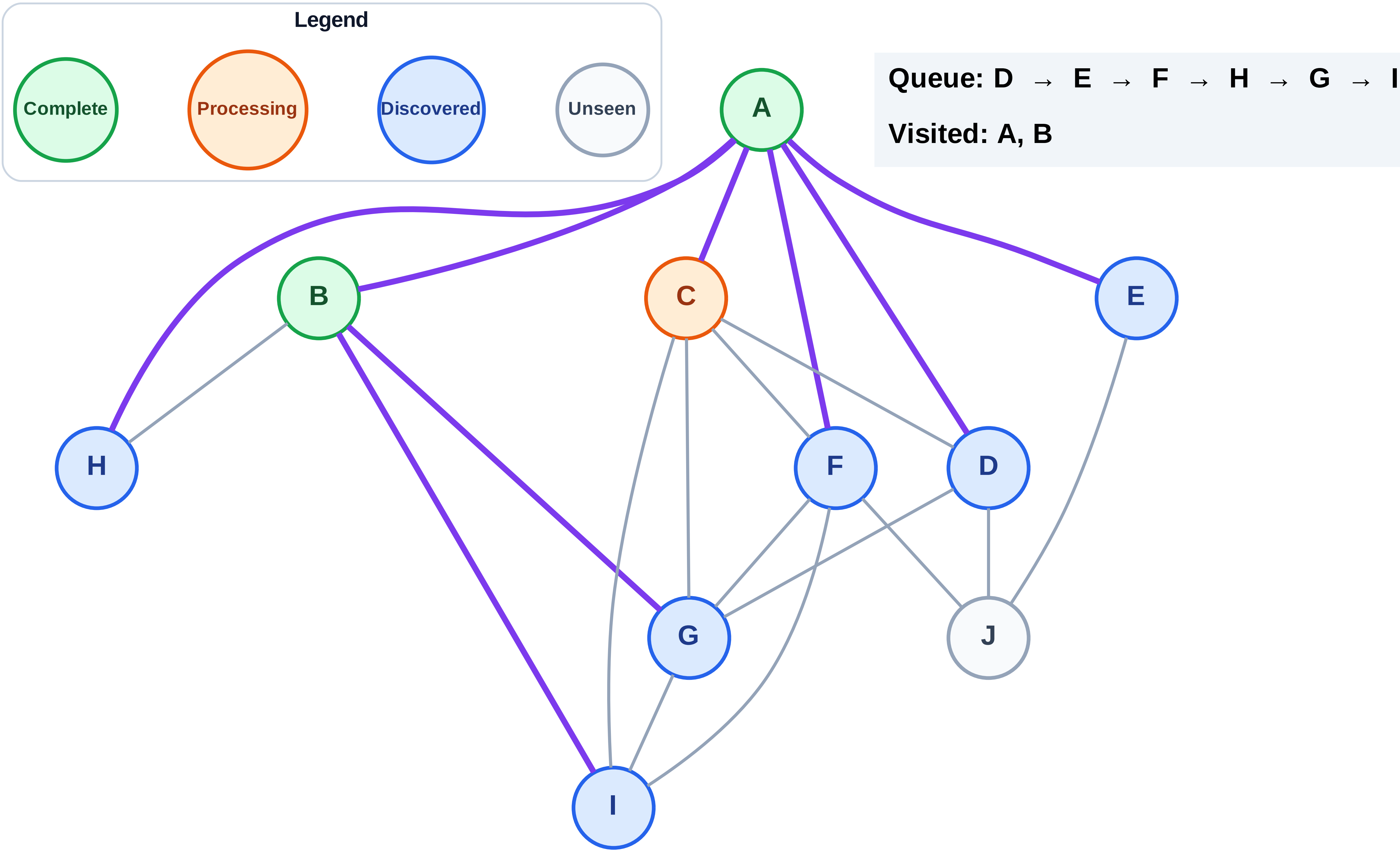


Queue: C → D → E → F → H → G → I

Visited: A, B

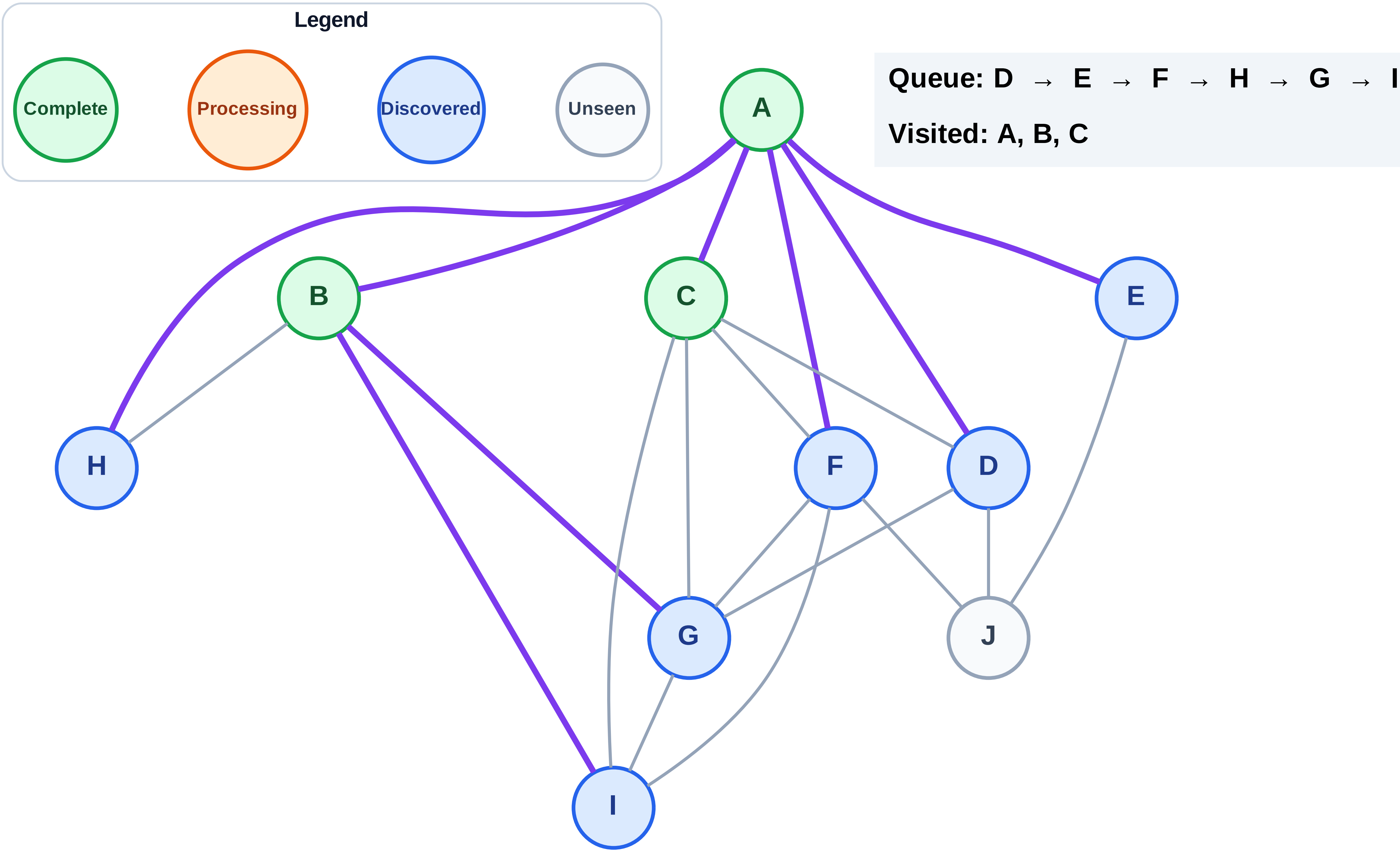
BFS Traversal

Step 6: Process node C



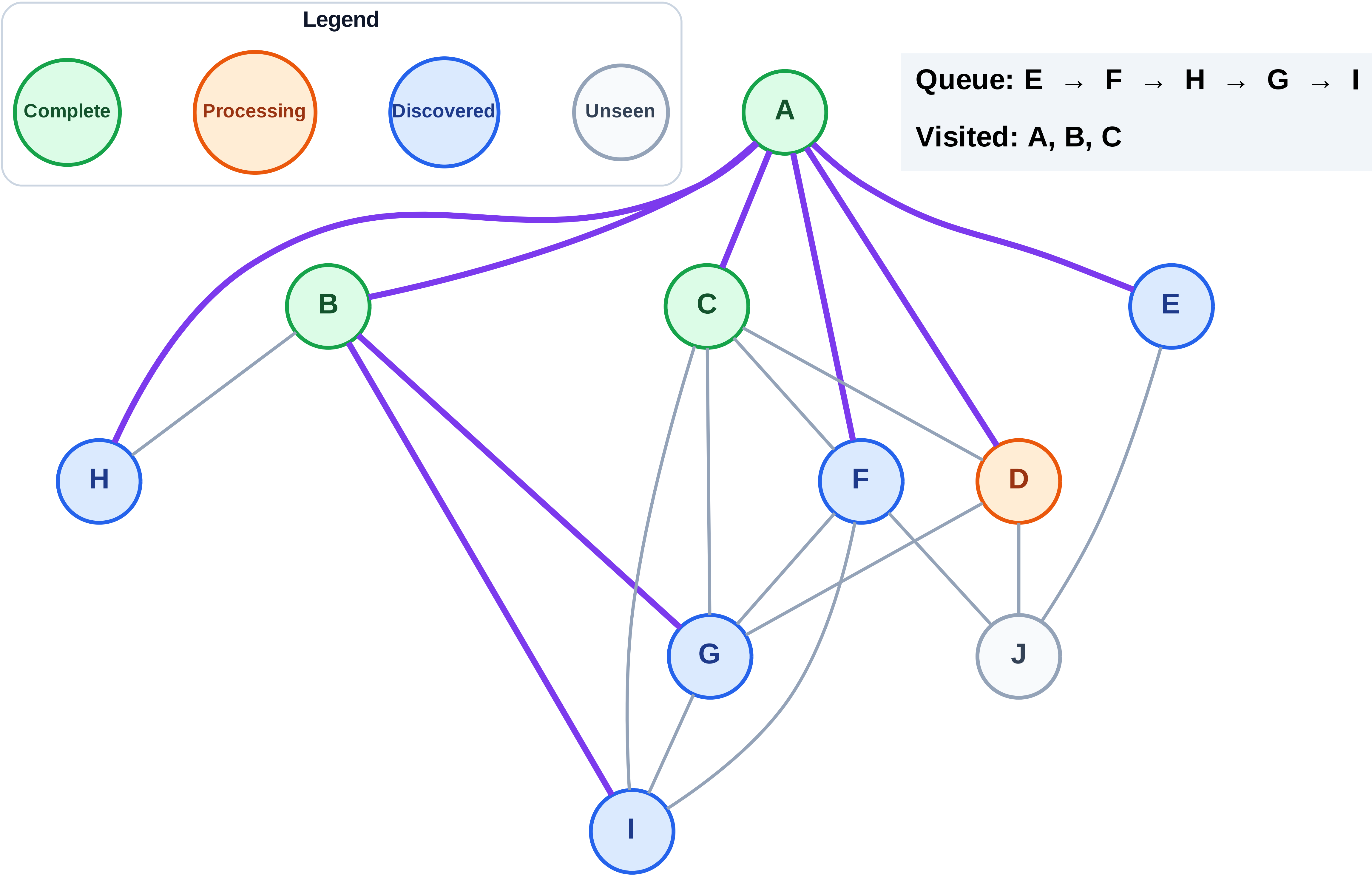
BFS Traversal

Step 7: No new neighbors from C



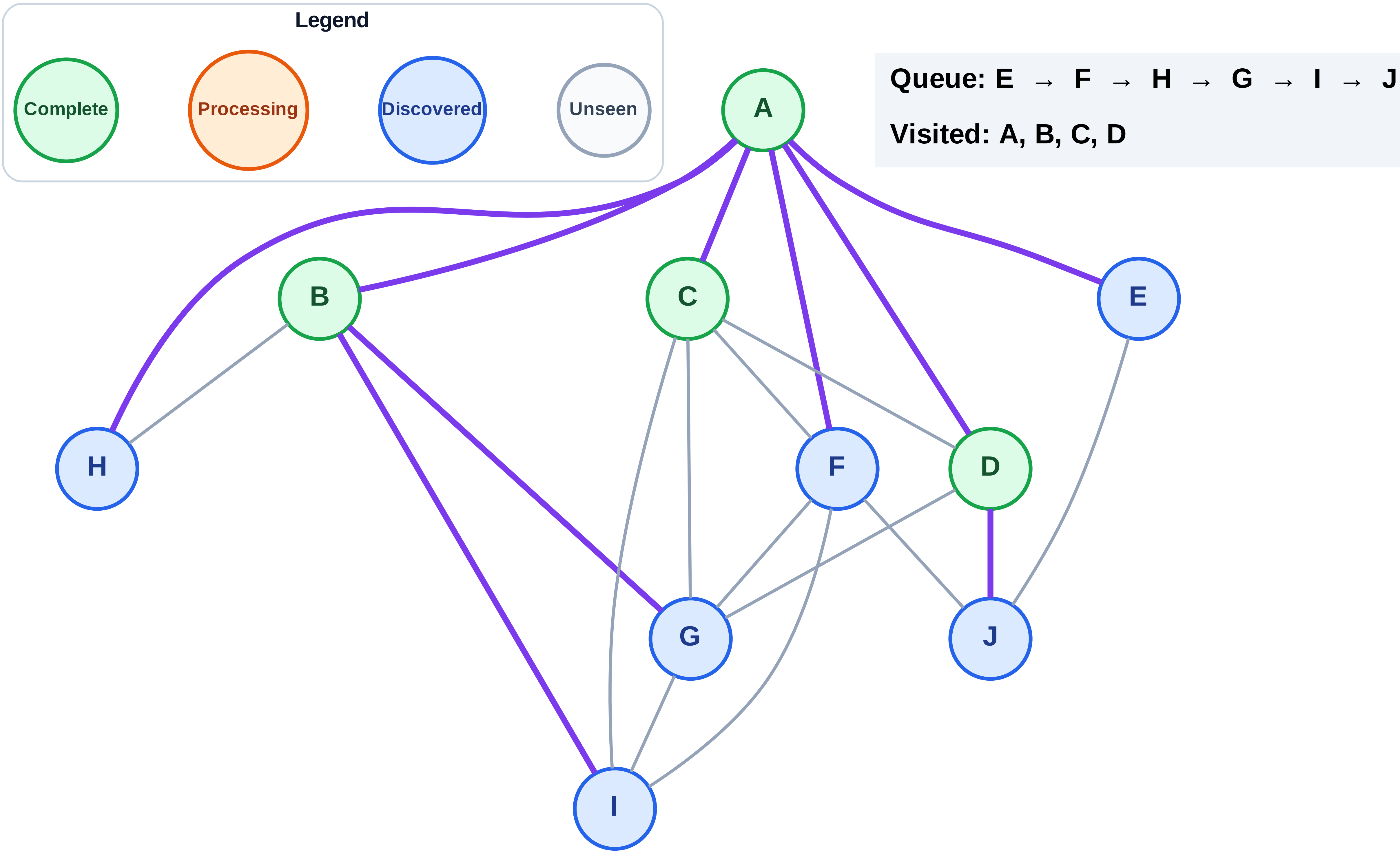
BFS Traversal

Step 8: Process node D



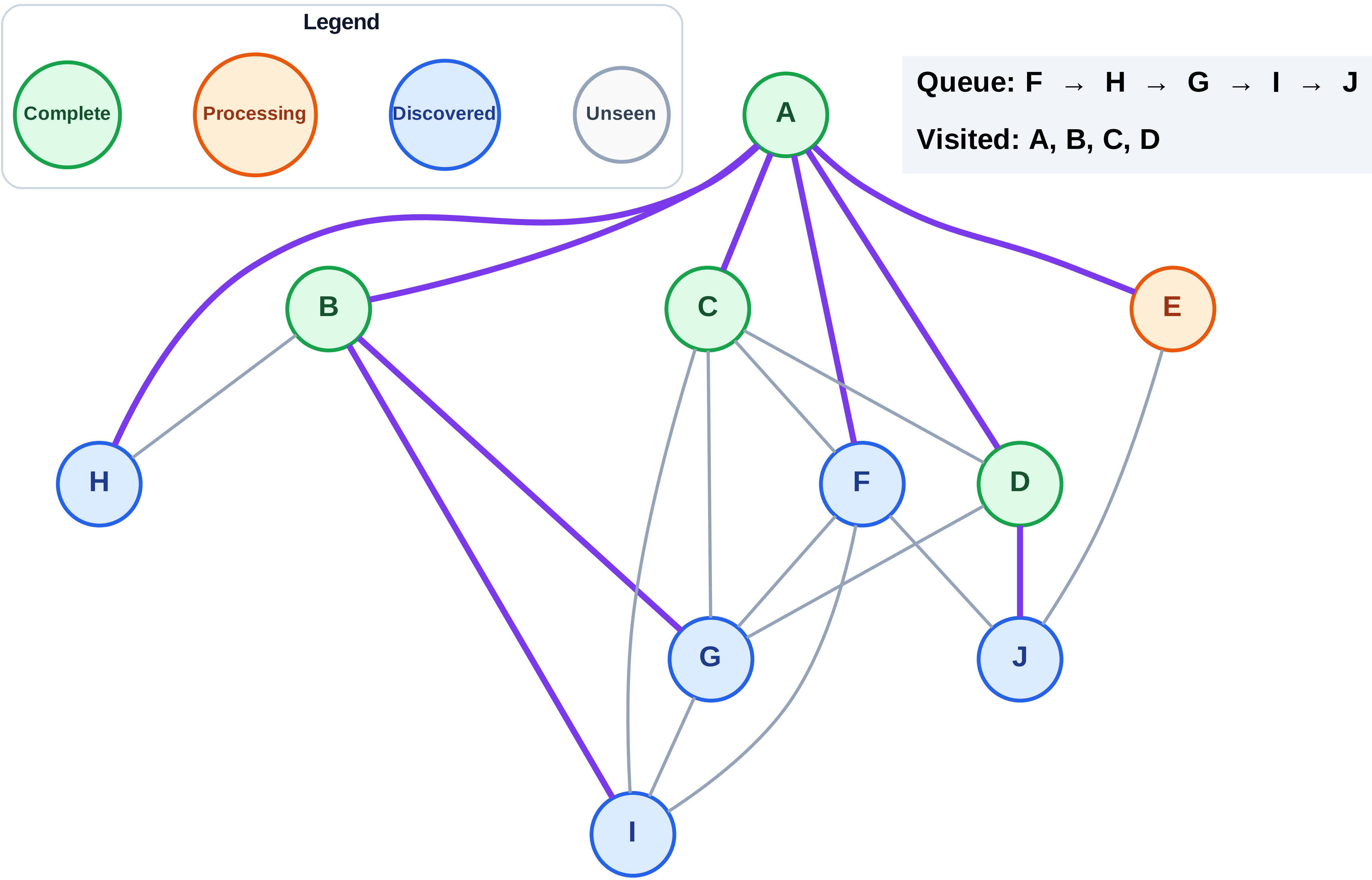
BFS Traversal

Step 9: Discover J from D



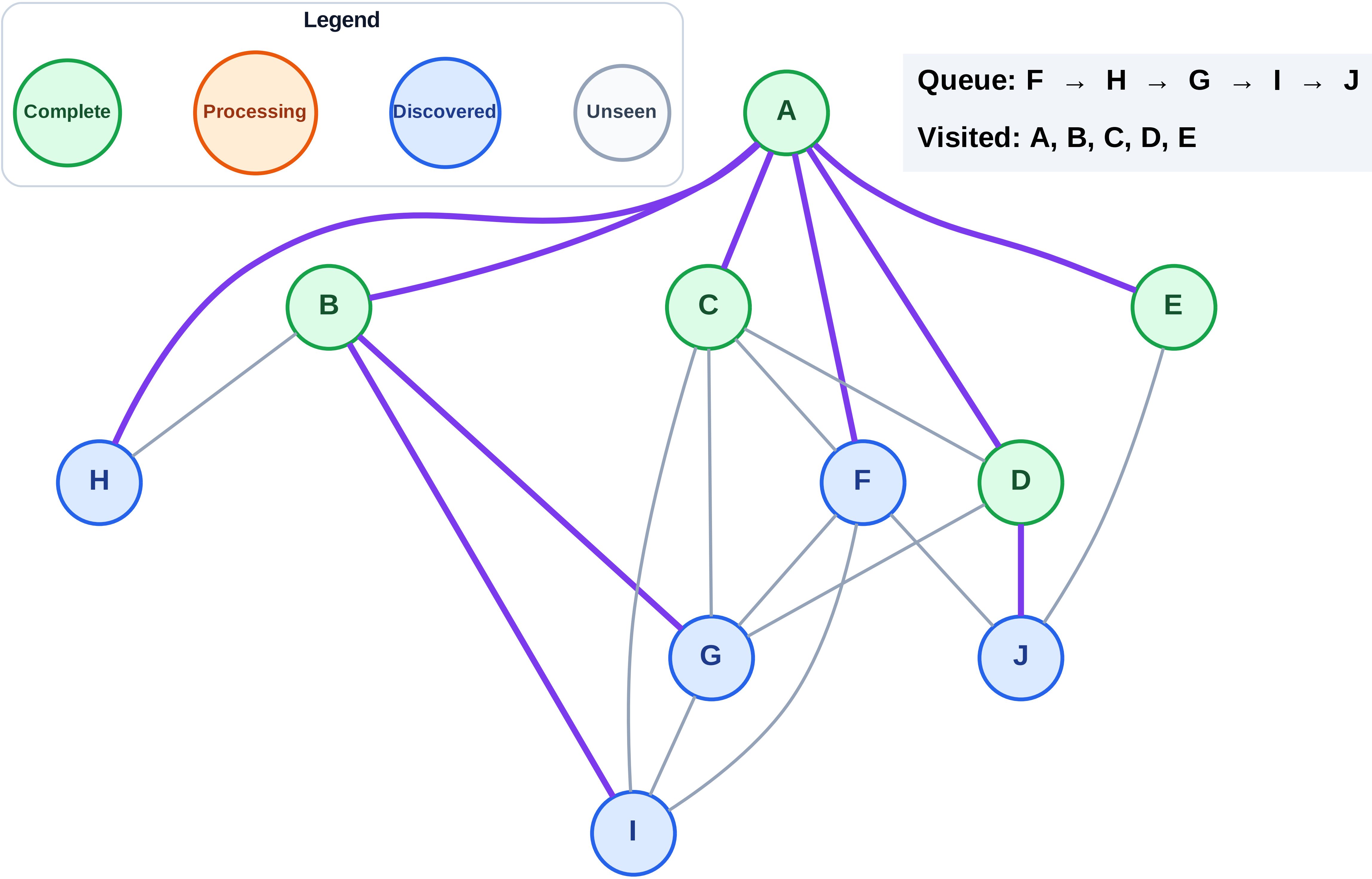
BFS Traversal

Step 10: Process node E



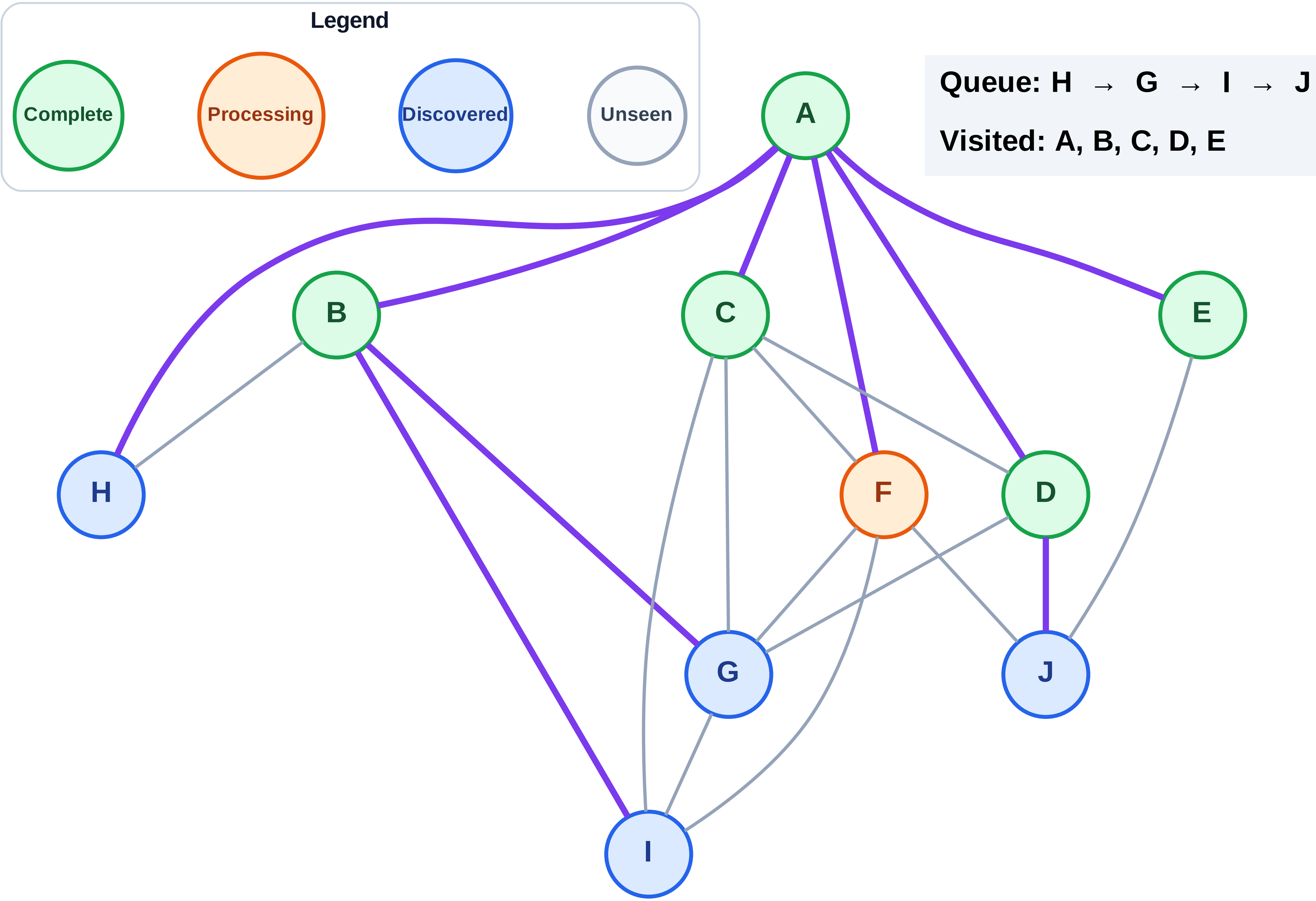
BFS Traversal

Step 11: No new neighbors from E



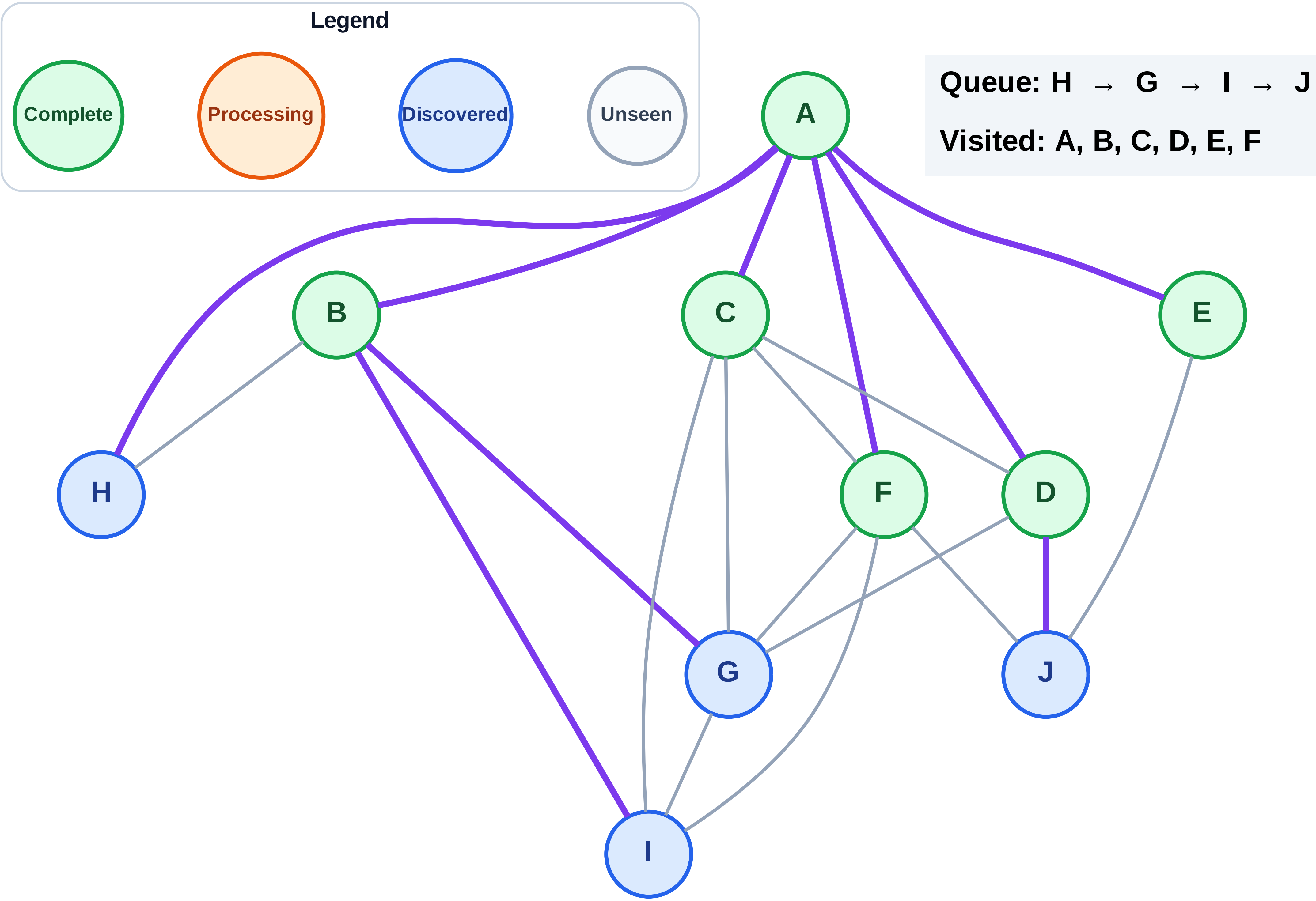
BFS Traversal

Step 12: Process node F



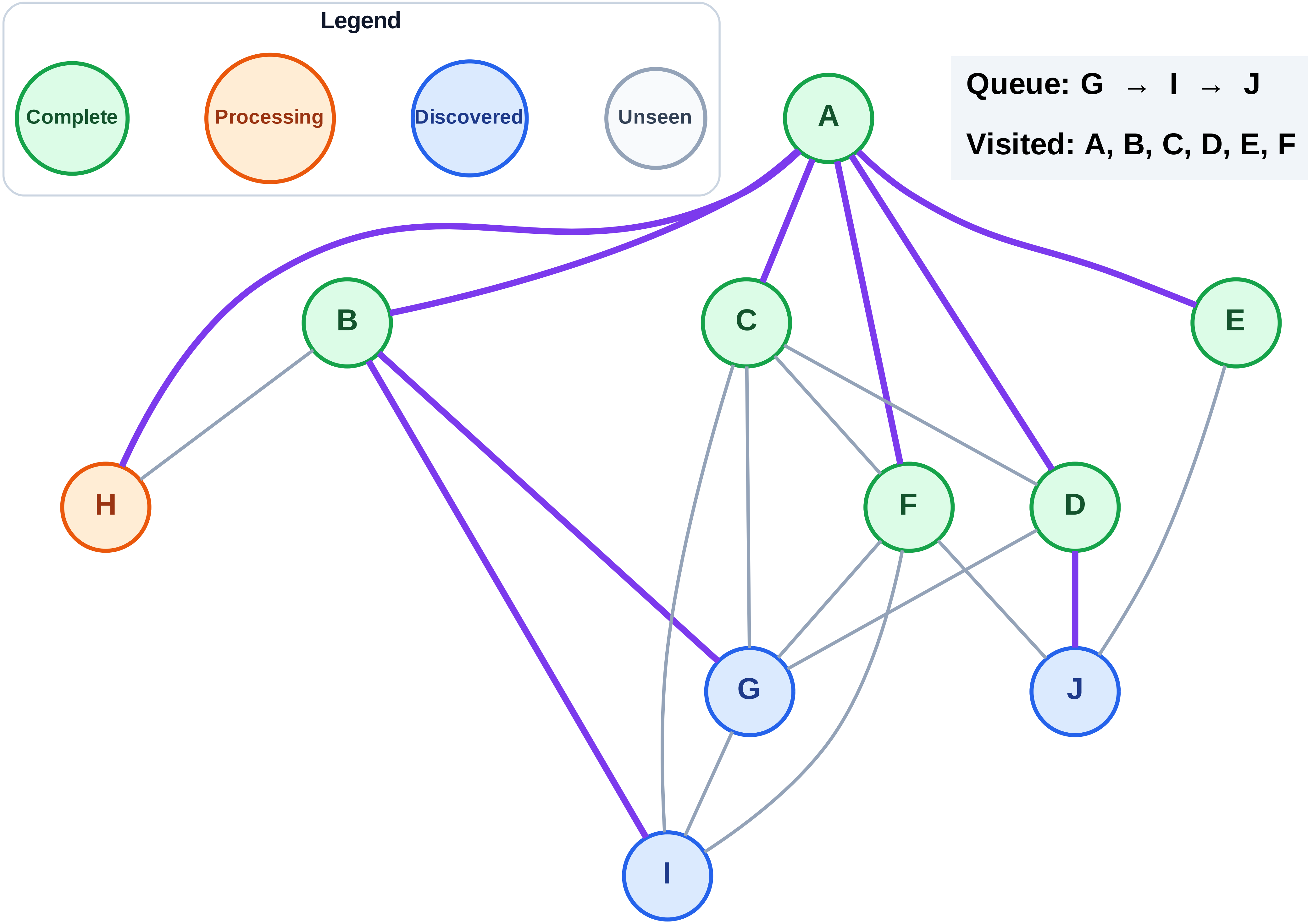
BFS Traversal

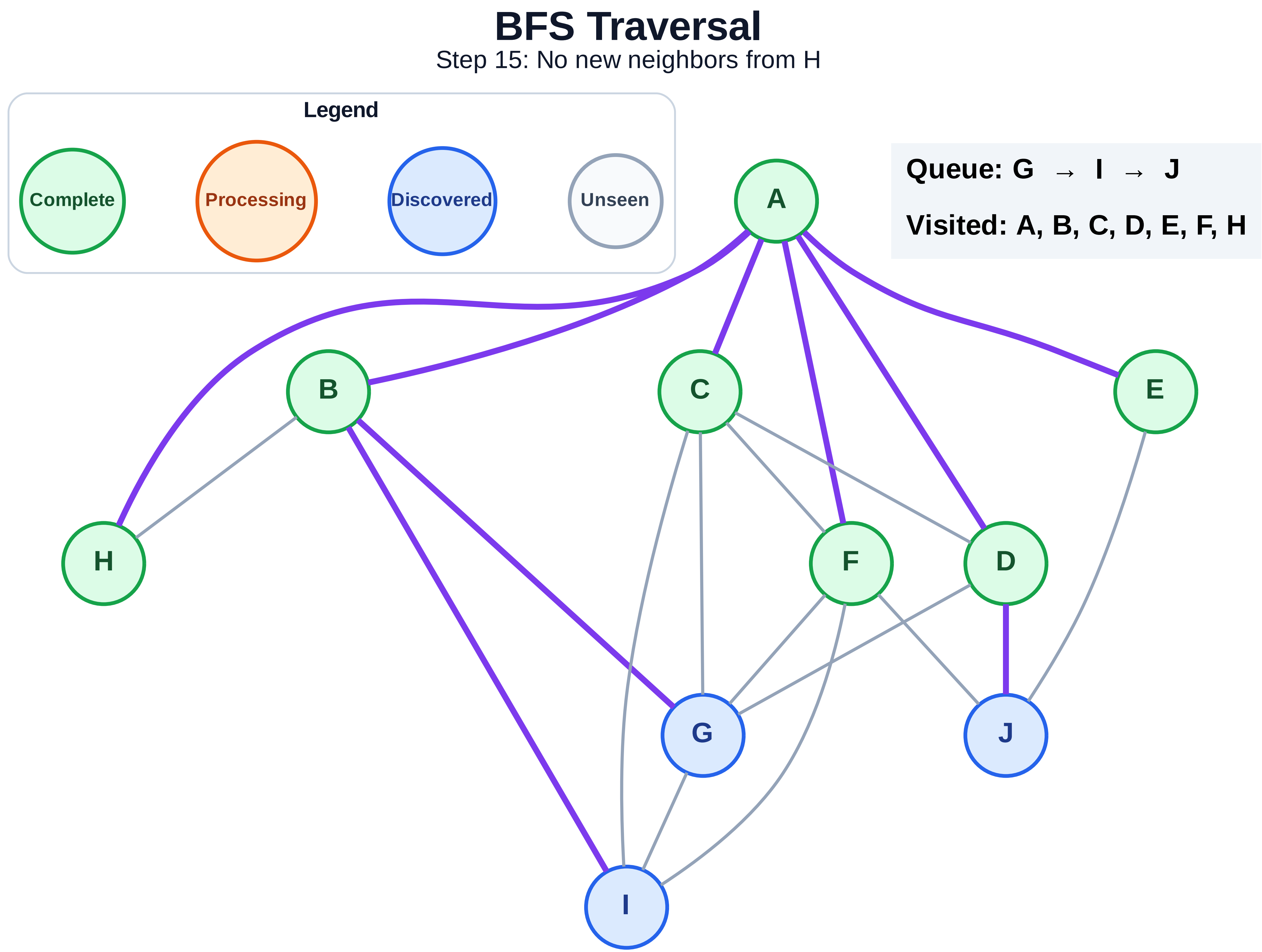
Step 13: No new neighbors from F



BFS Traversal

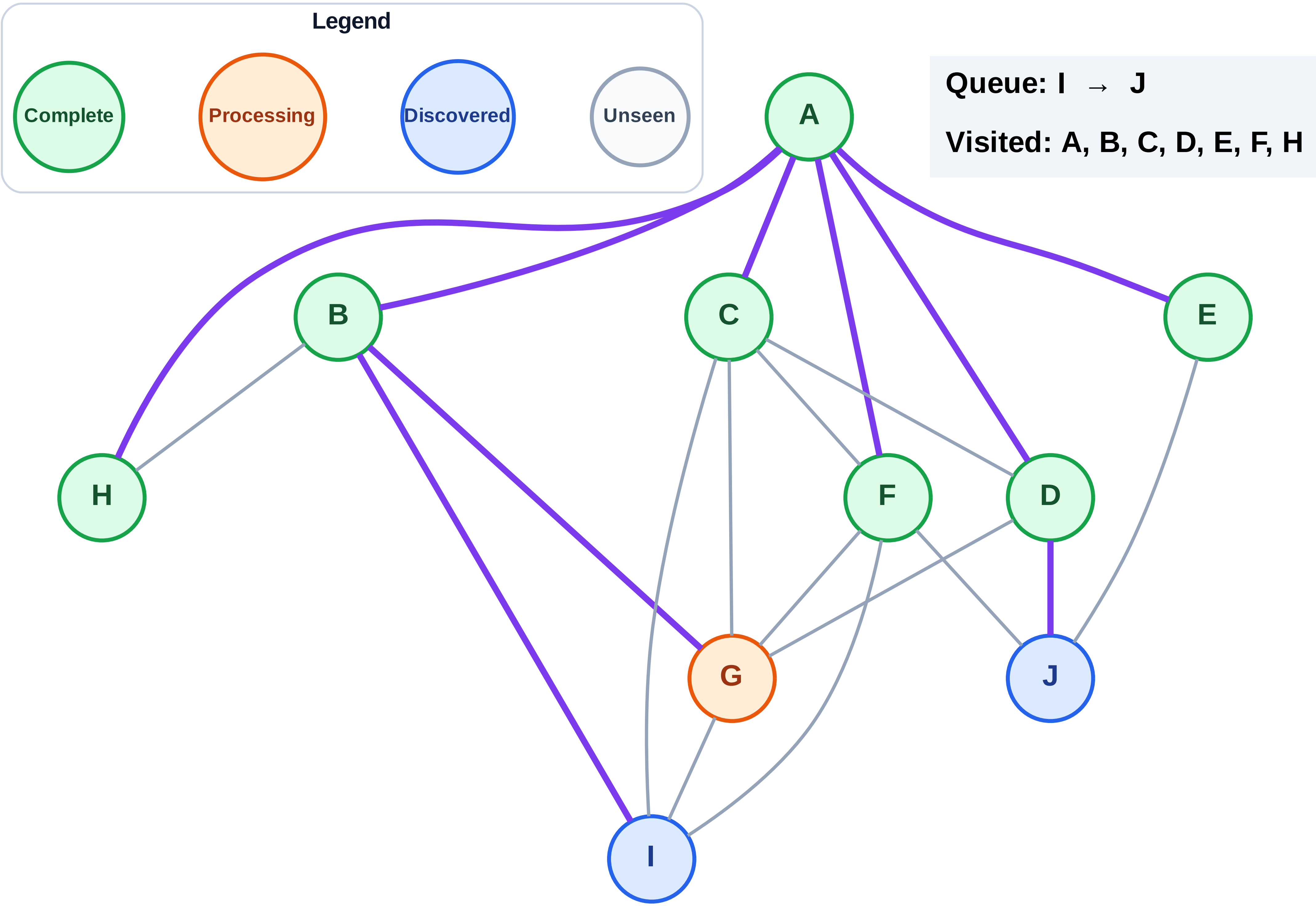
Step 14: Process node H




BFS Traversal
Step 15: No new neighbors from H
Legend
Complete
Processing
Discovered
Unseen
Queue: G → I → J
Visited: A, B, C, D, E, F, H
A
B
C
E
H
F
D
G
J
I

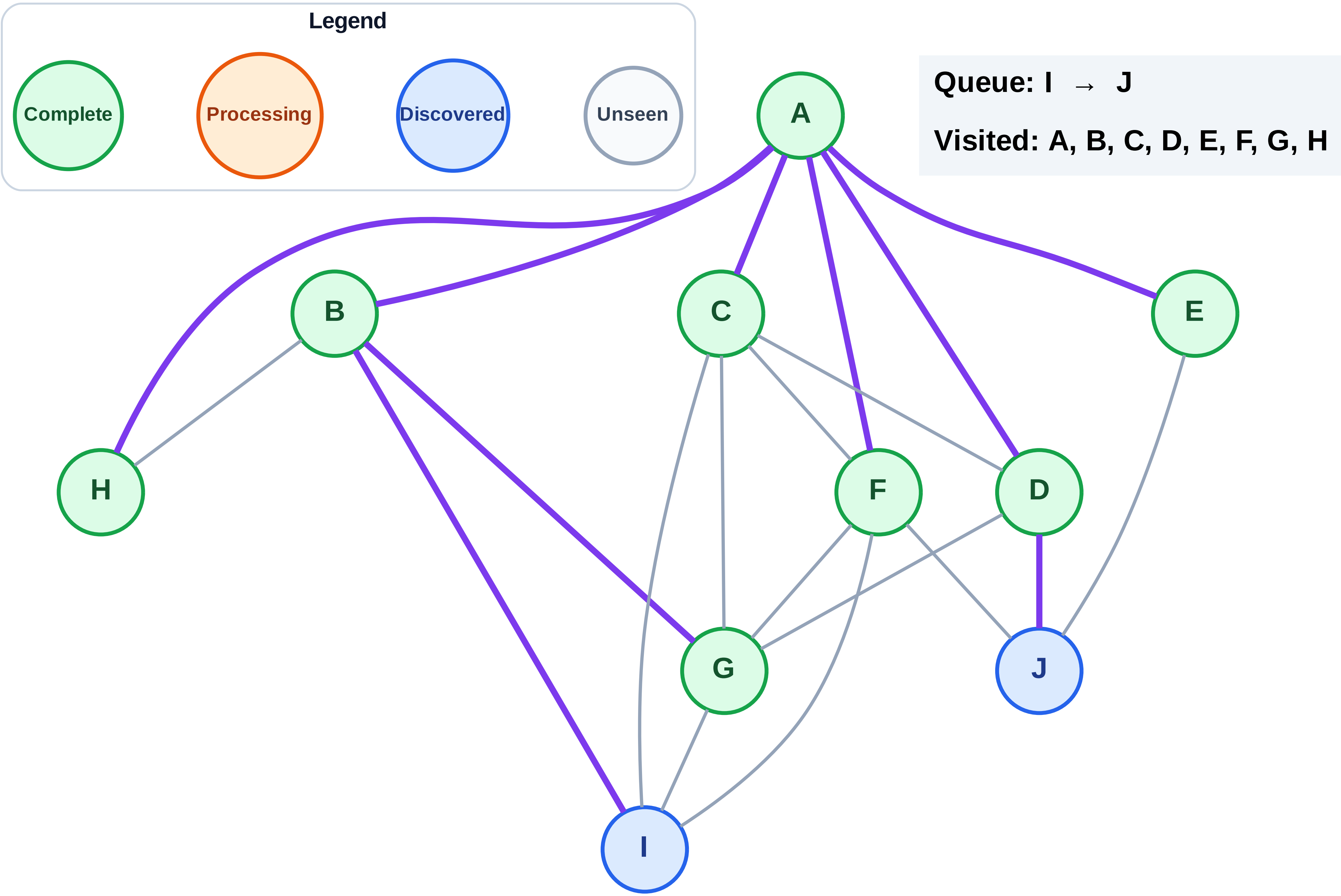
BFS Traversal

Step 16: Process node G



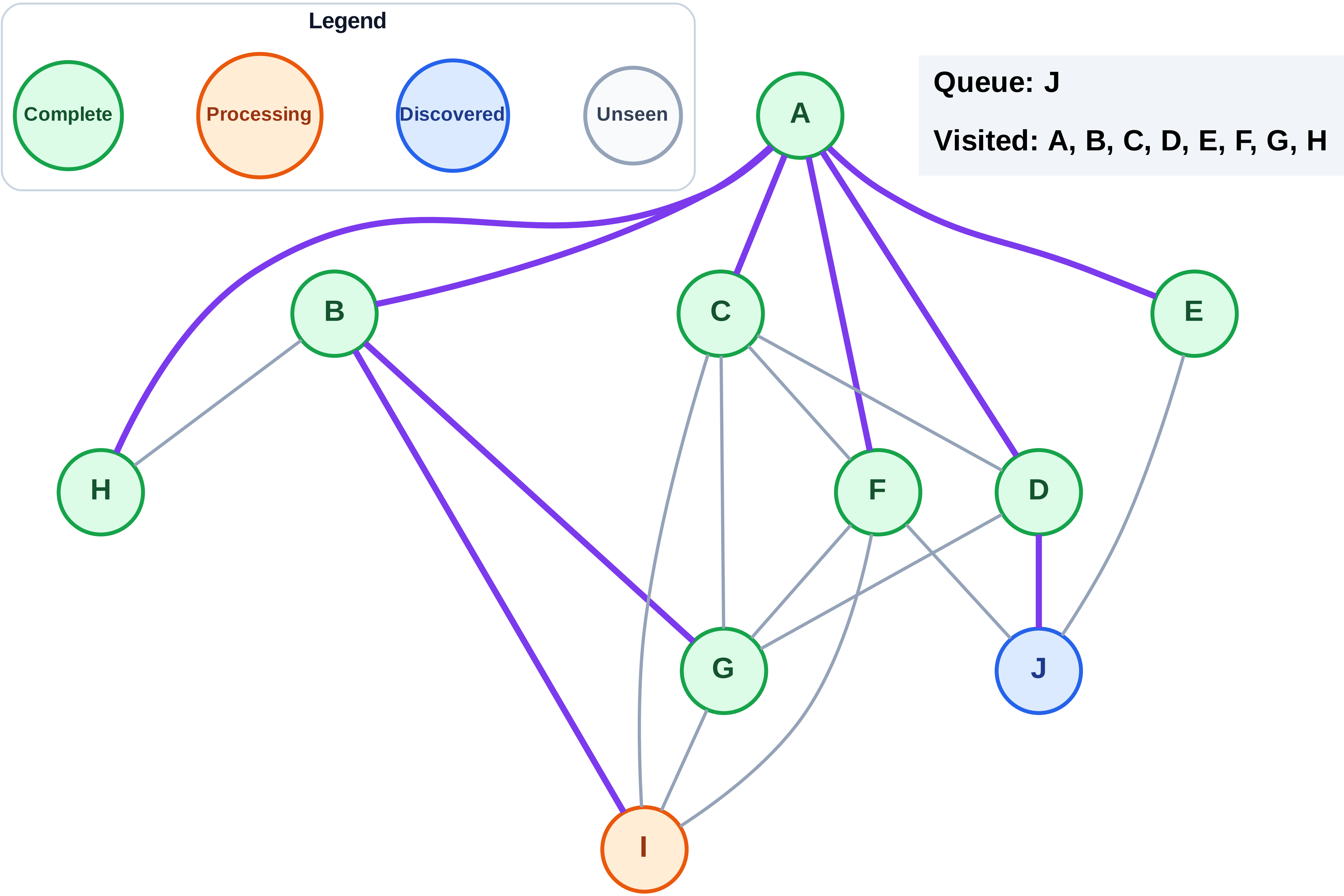
BFS Traversal

Step 17: No new neighbors from G



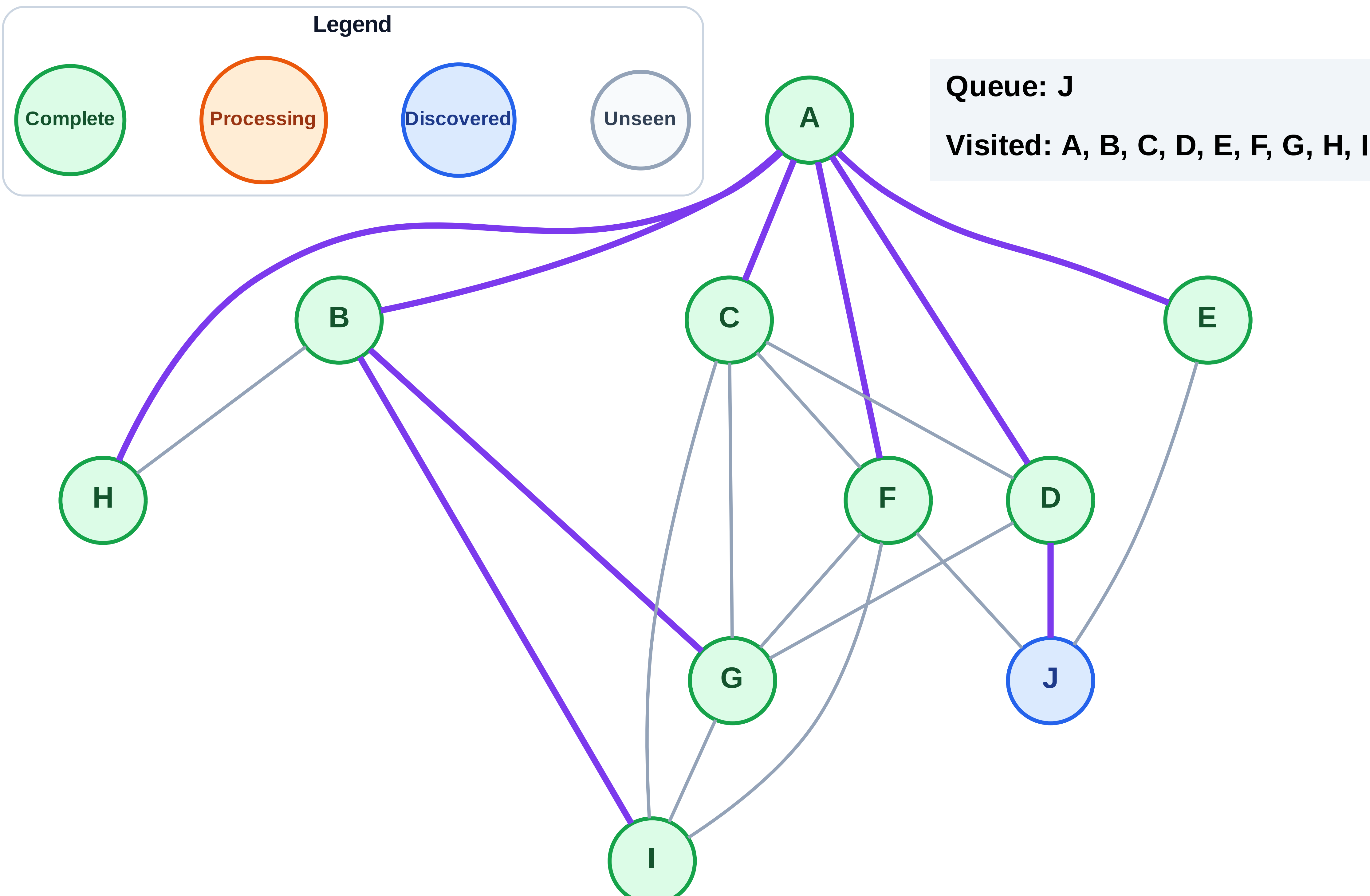
BFS Traversal

Step 18: Process node I



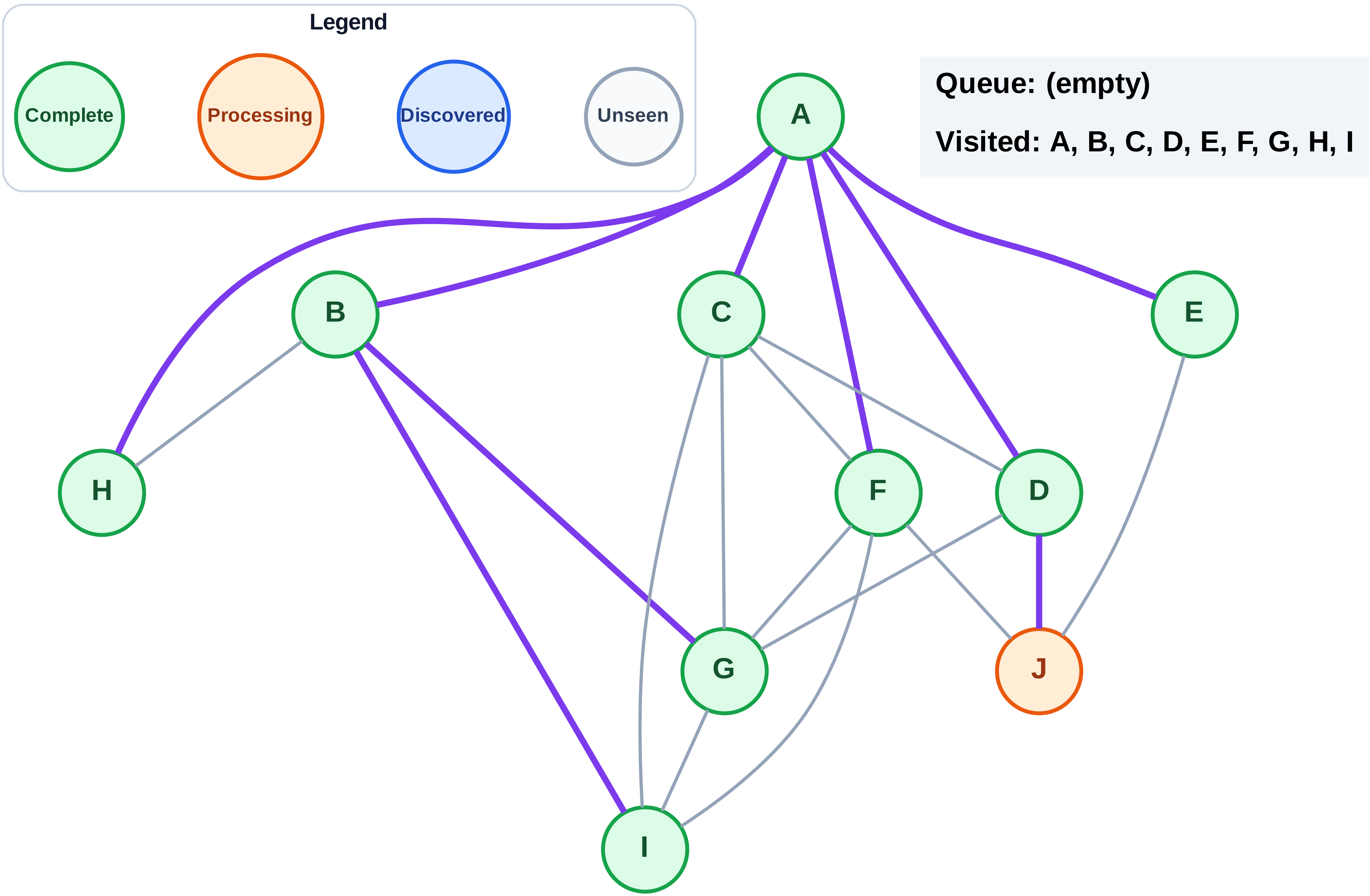
BFS Traversal

Step 19: No new neighbors from I



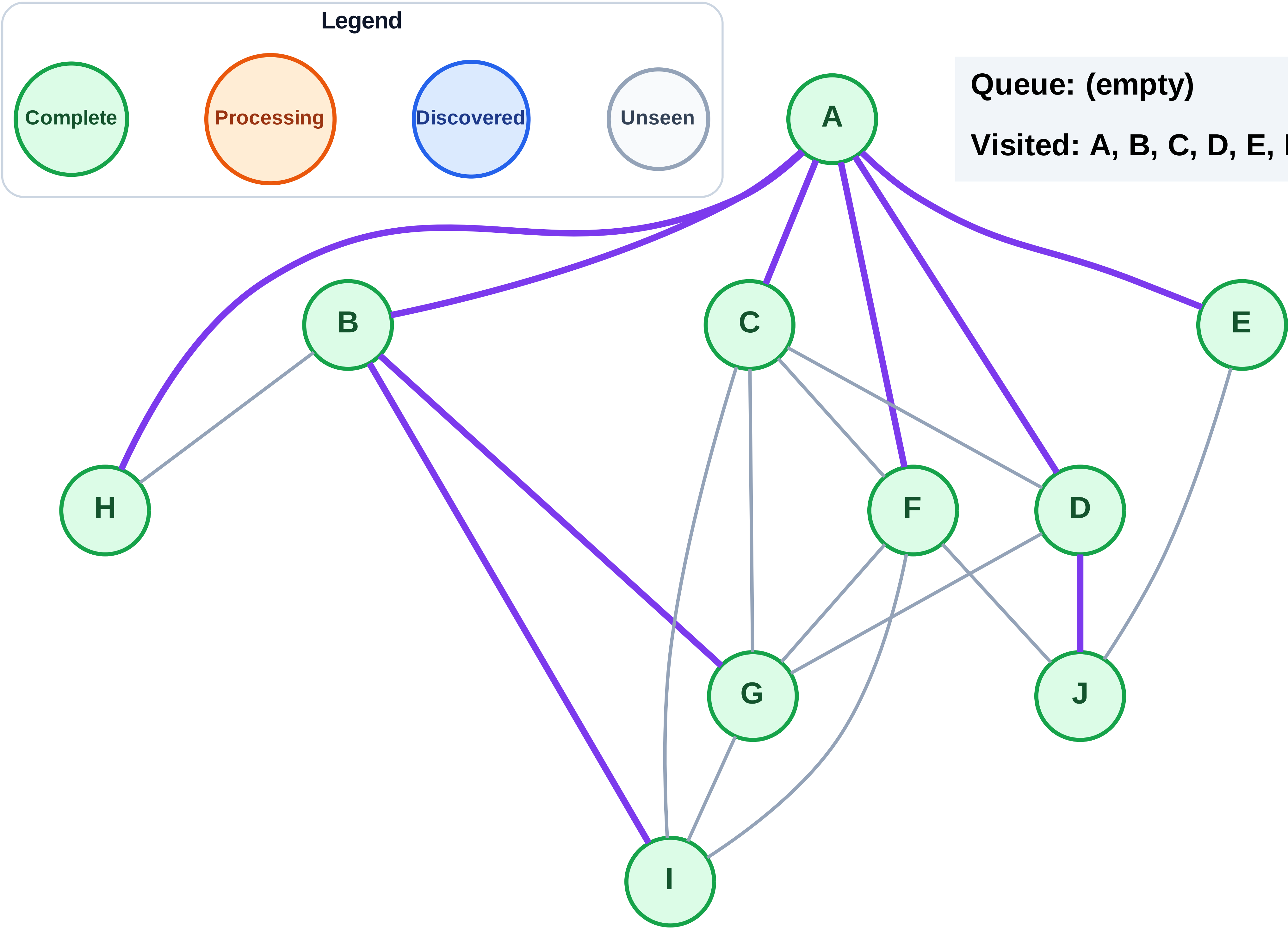
BFS Traversal

Step 20: Process node J



BFS Traversal

Step 21: No new neighbors from J

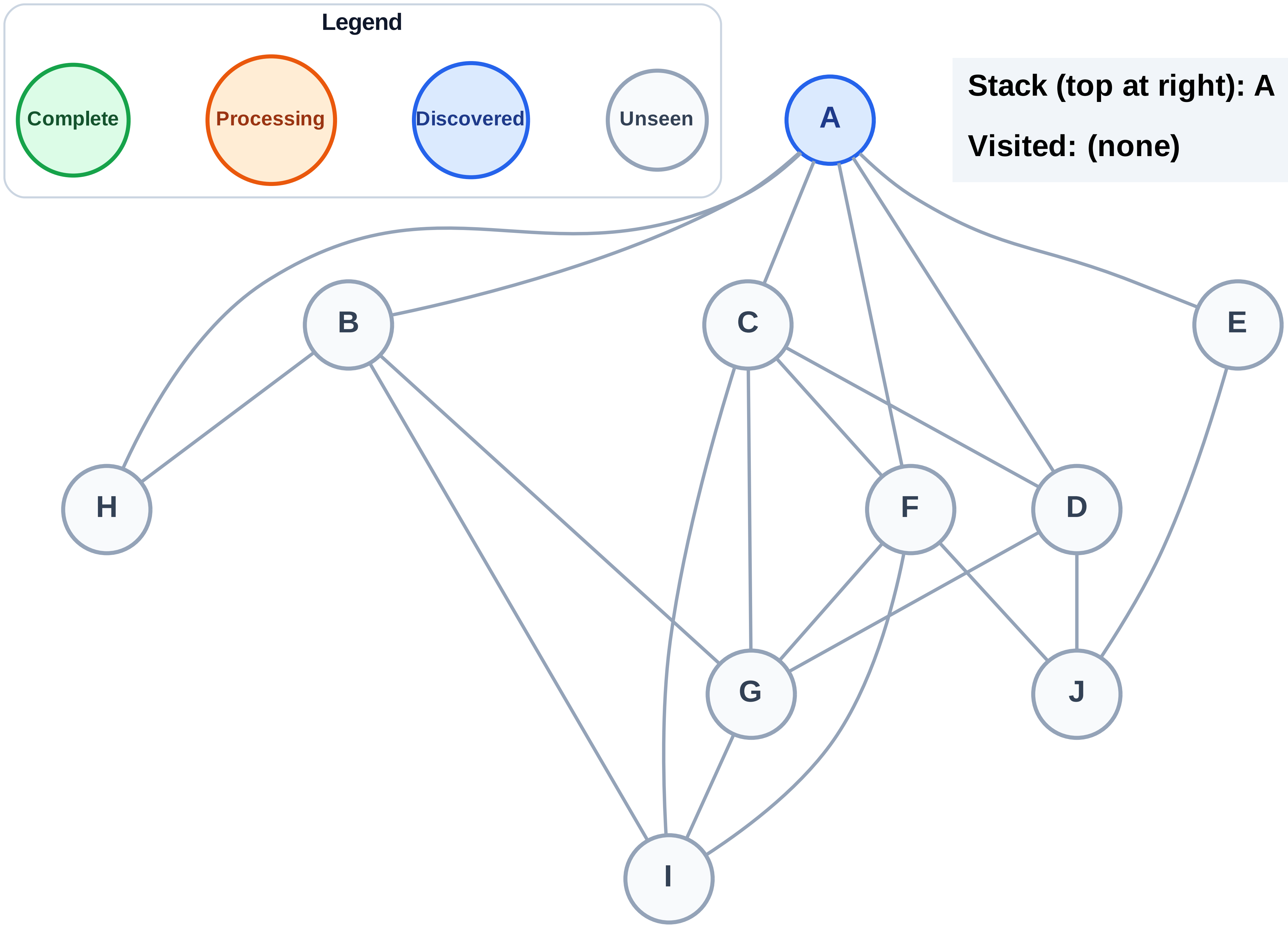


Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J

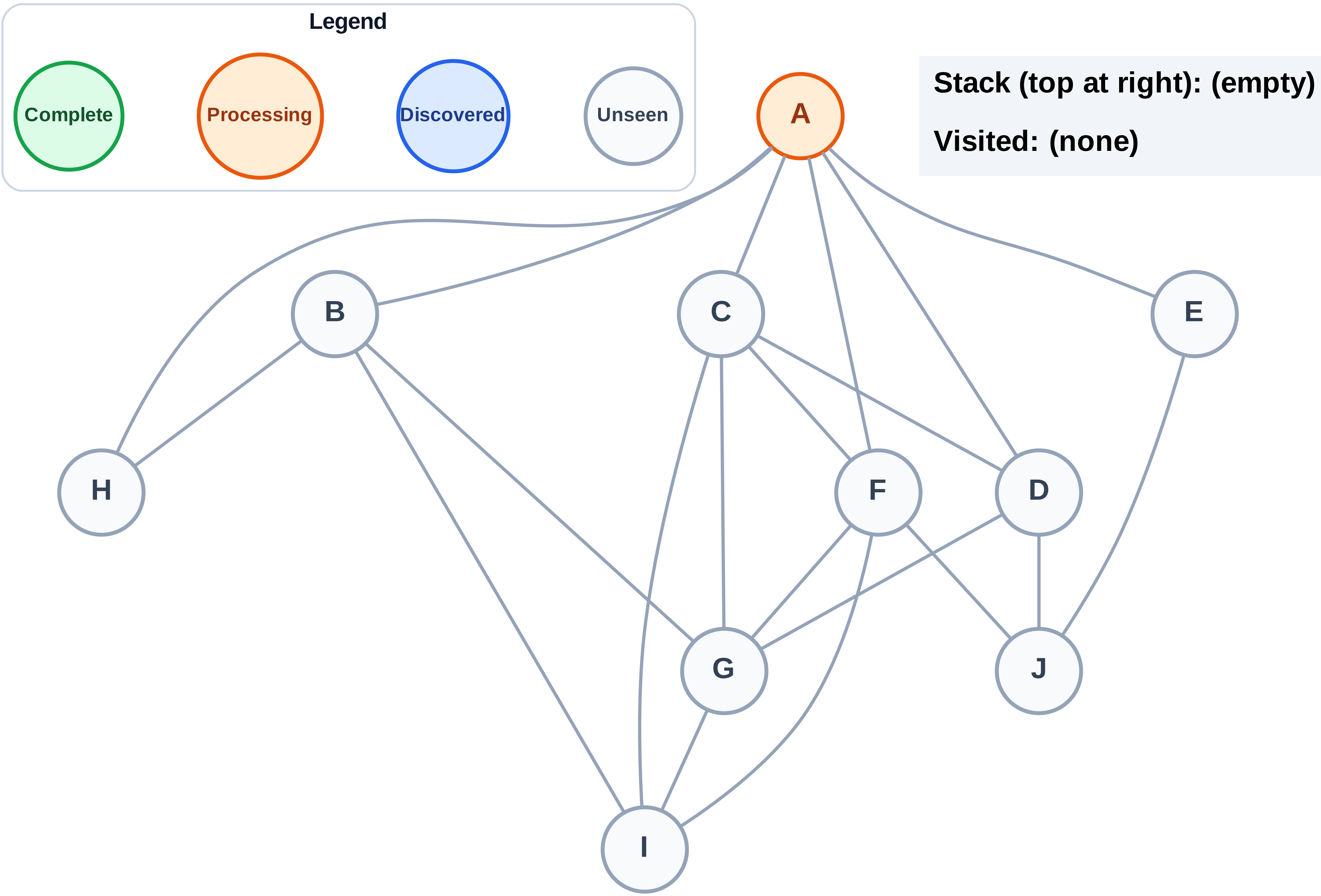
DFS Traversal

Step 1: Start at A



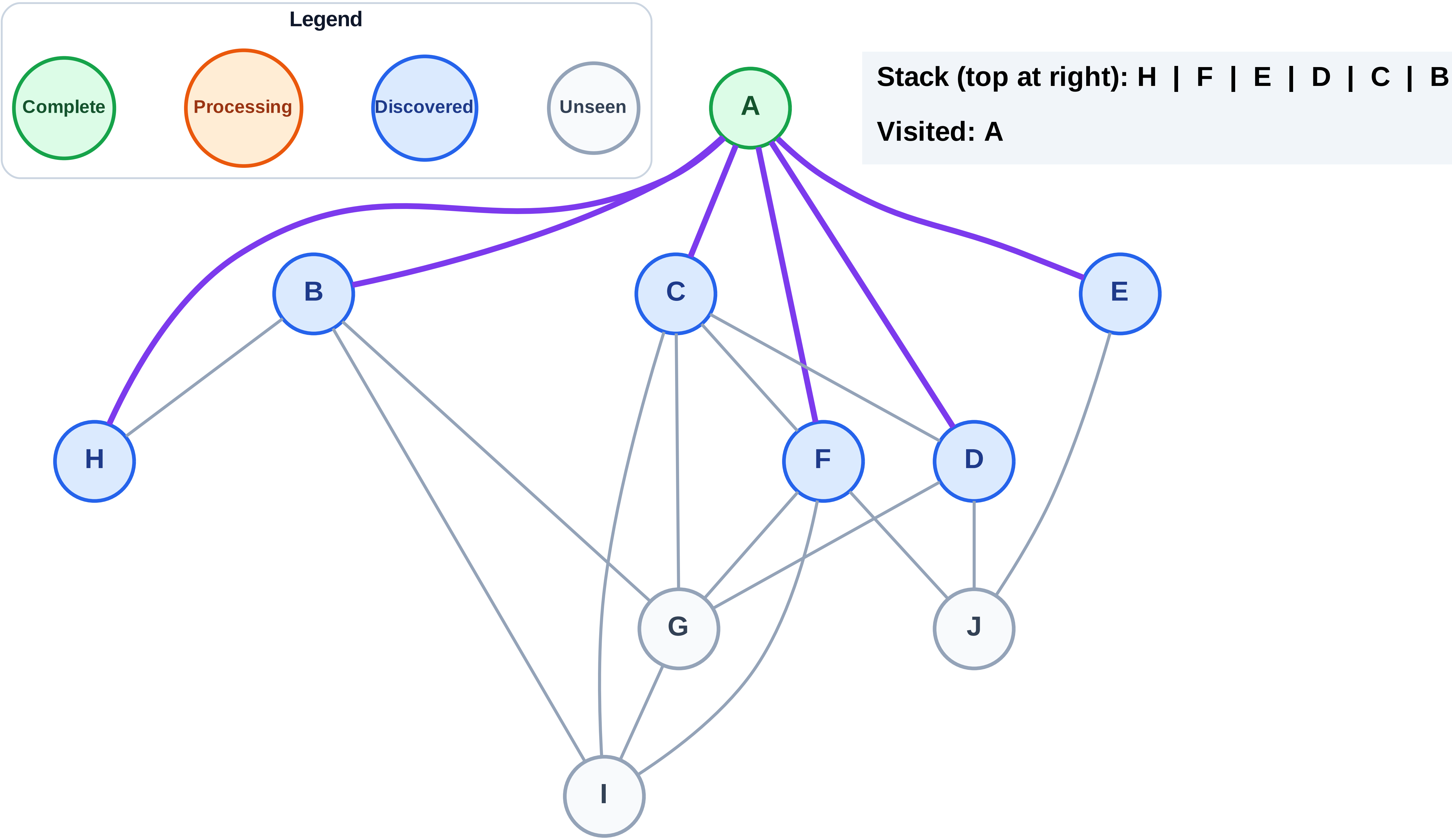
DFS Traversal

Step 2: Process node A



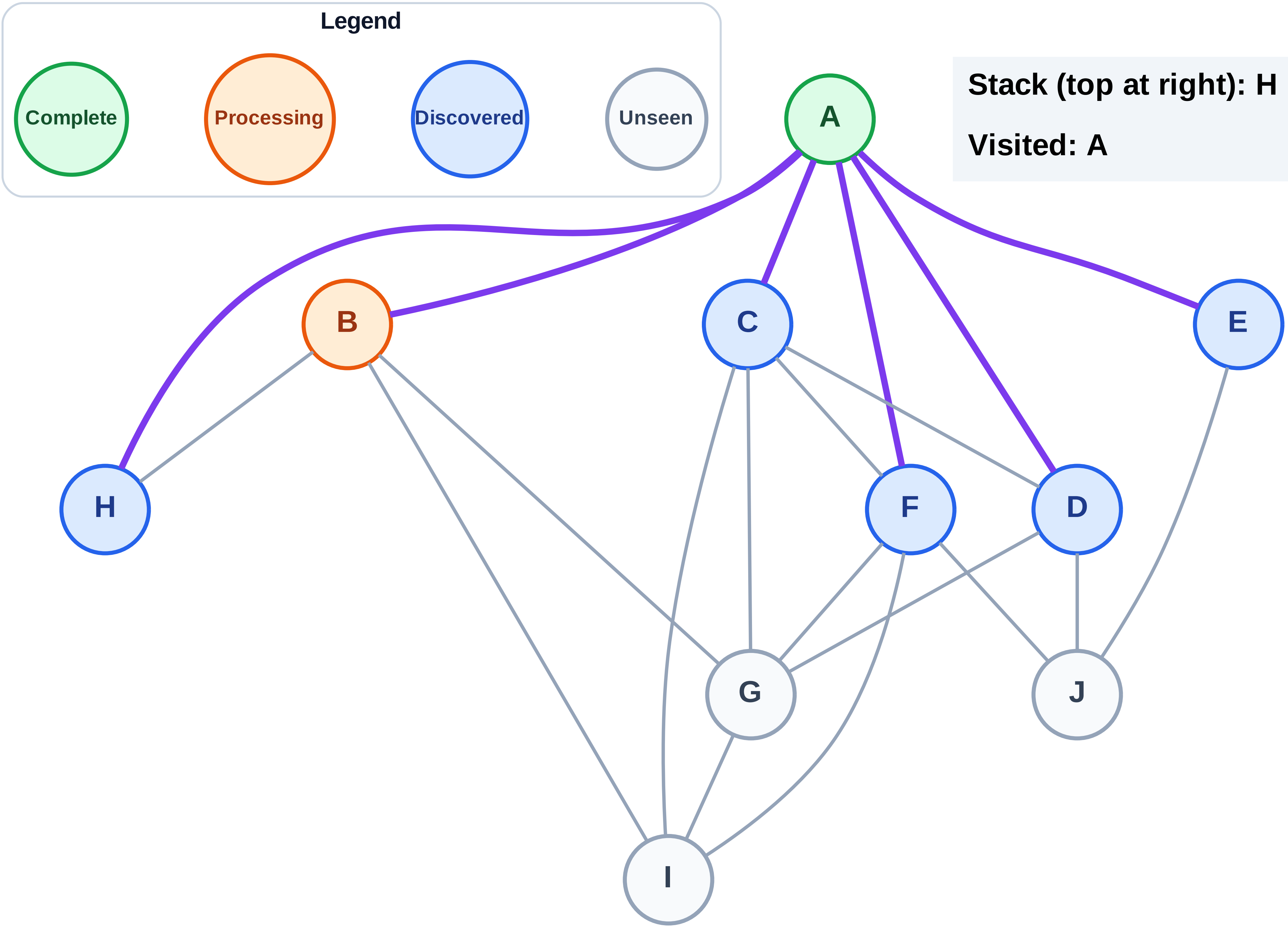
DFS Traversal

Step 3: Discover H, F, E, D, C, B from A



DFS Traversal

Step 4: Process node B

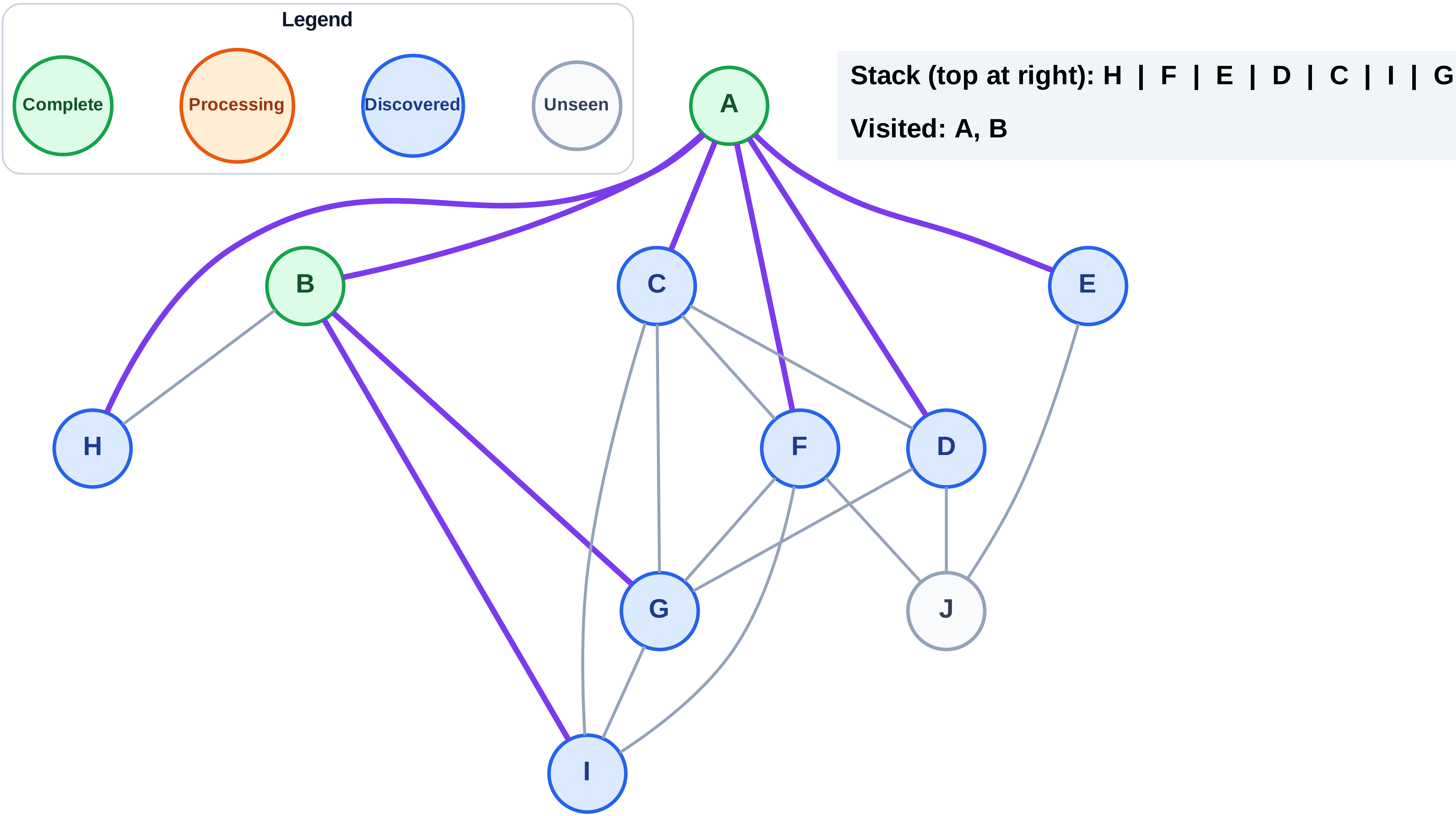


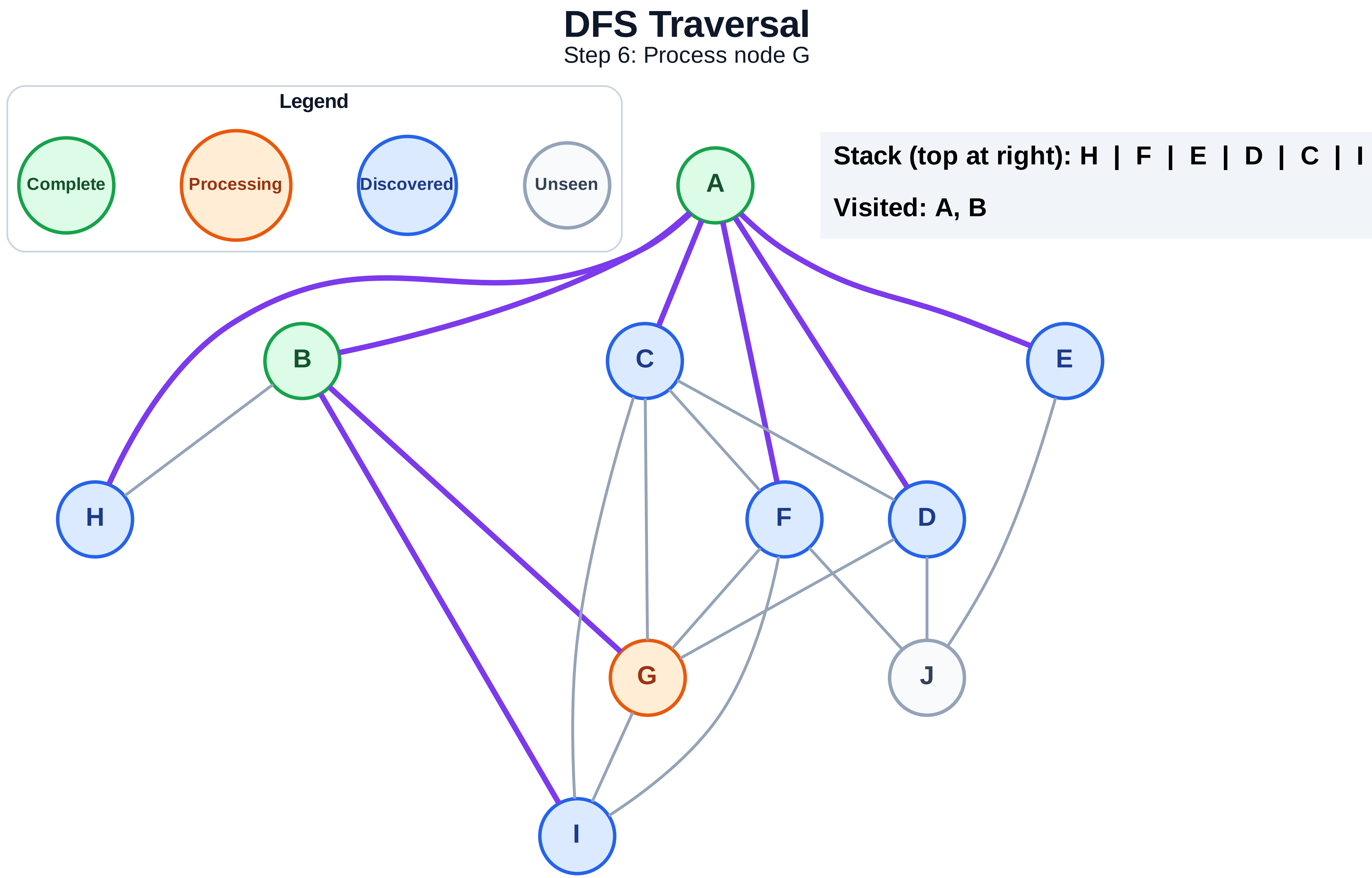
Stack (top at right): H | F | E | D | C

Visited: A

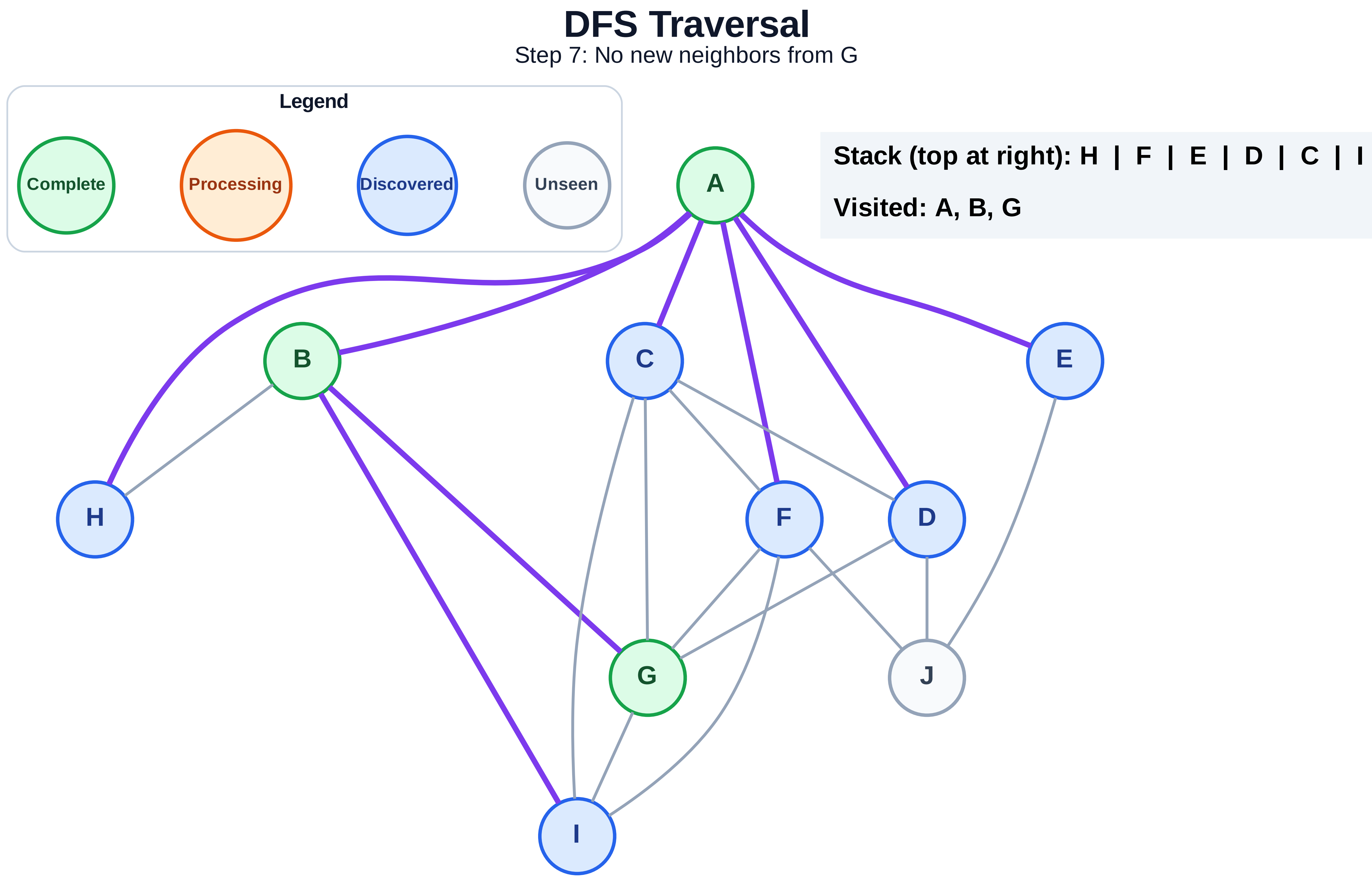
DFS Traversal

Step 5: Discover I, G from B





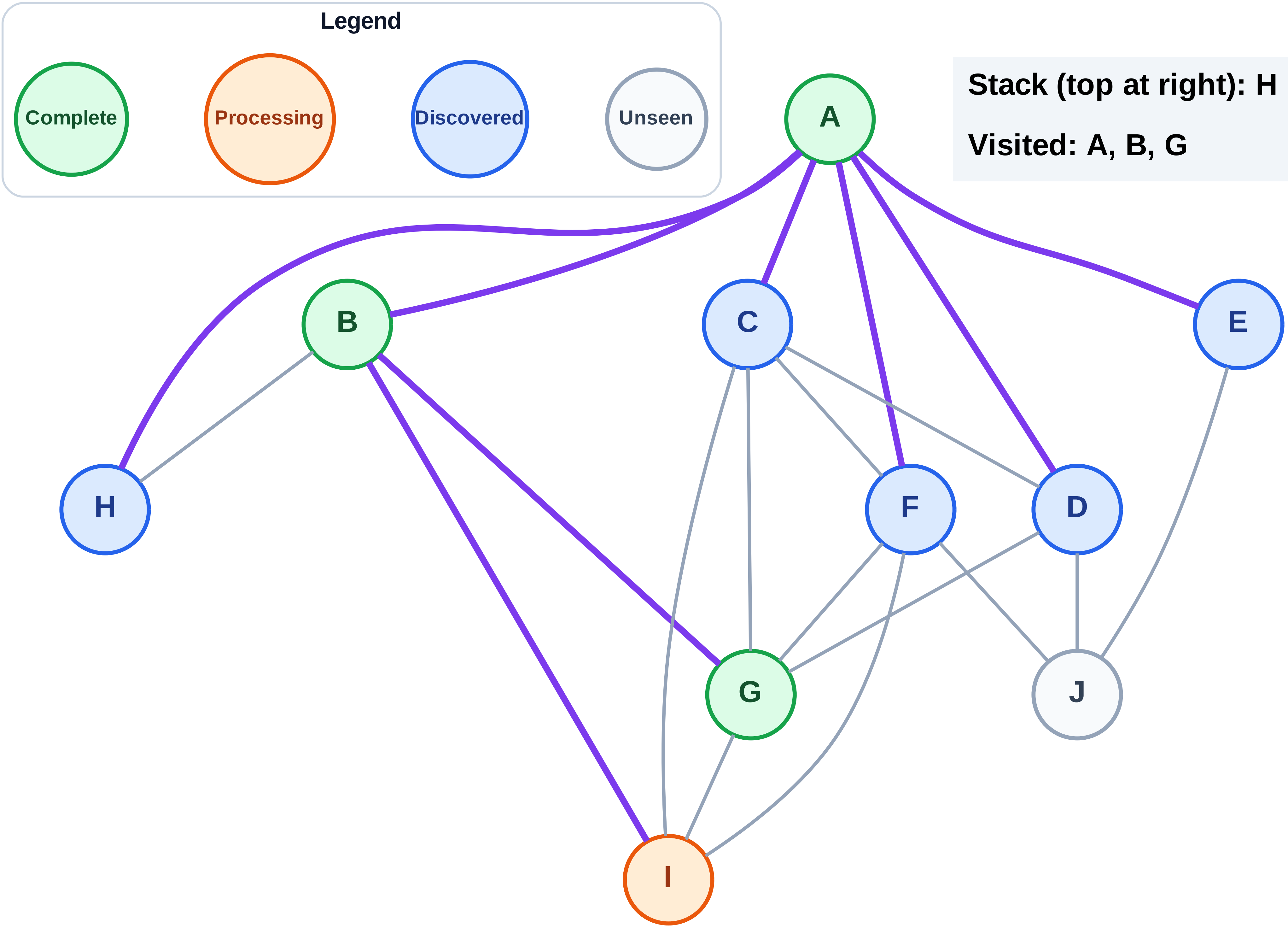
DFS Traversal
Step 6: Process node G
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): H | F | E | D | C | I
Visited: A, B
A
B
C
D
E
F
G
H
I
J



DFS Traversal
Step 7: No new neighbors from G
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): H | F | E | D | C | I
Visited: A, B, G
A
B
C
D
E
F
G
H
I
J

DFS Traversal

Step 8: Process node I

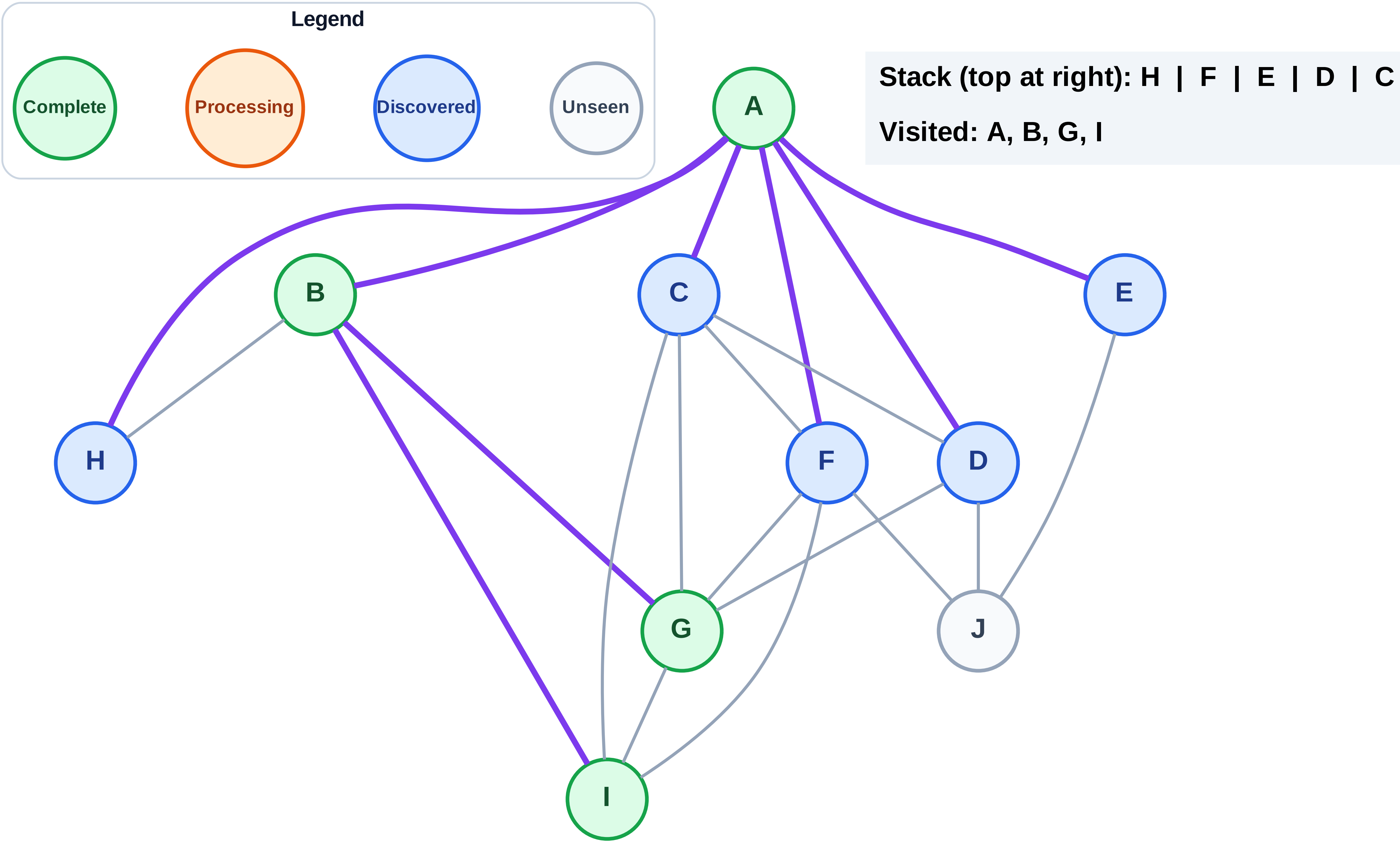


Stack (top at right): H | F | E | D | C

Visited: A, B, G

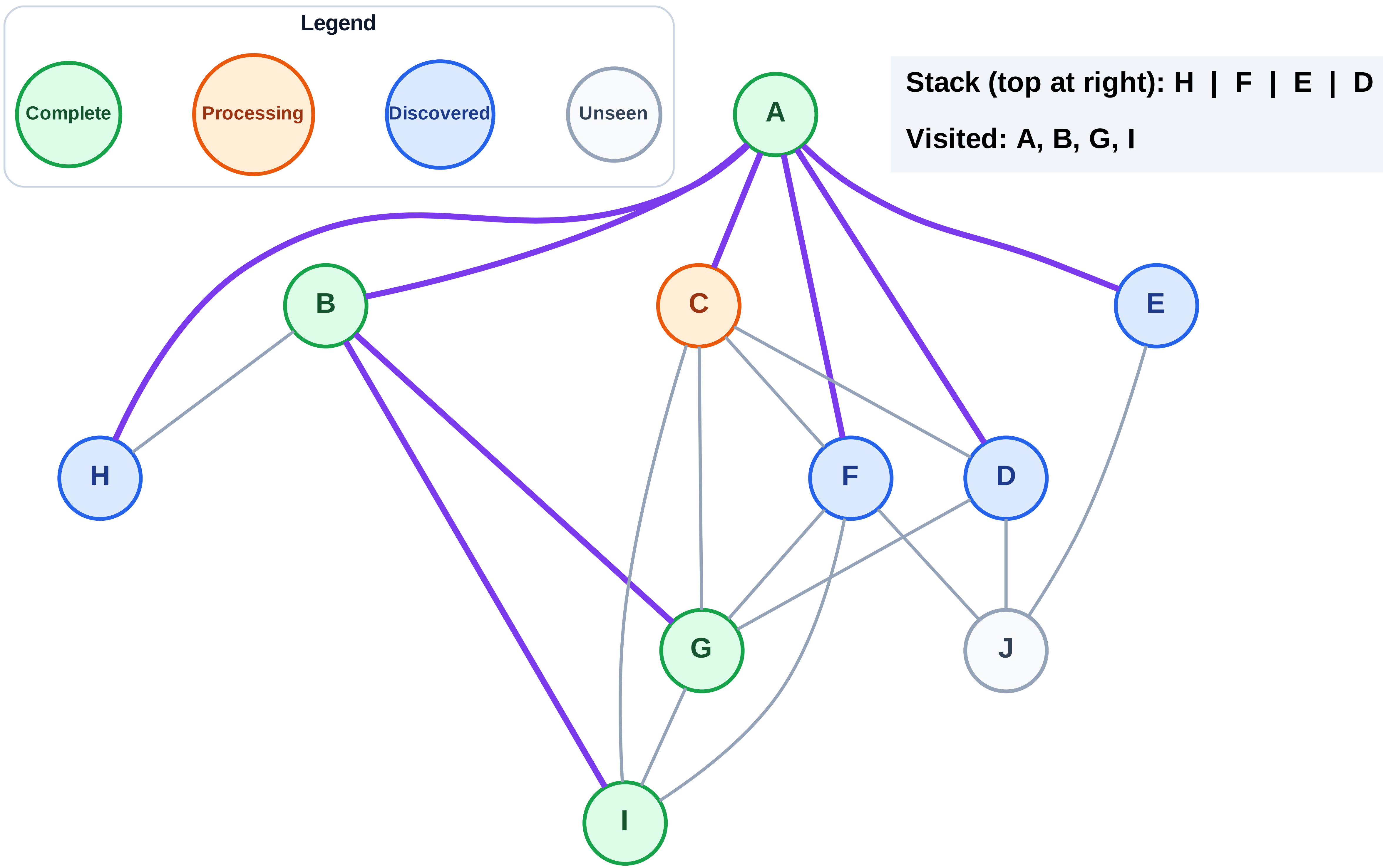
DFS Traversal

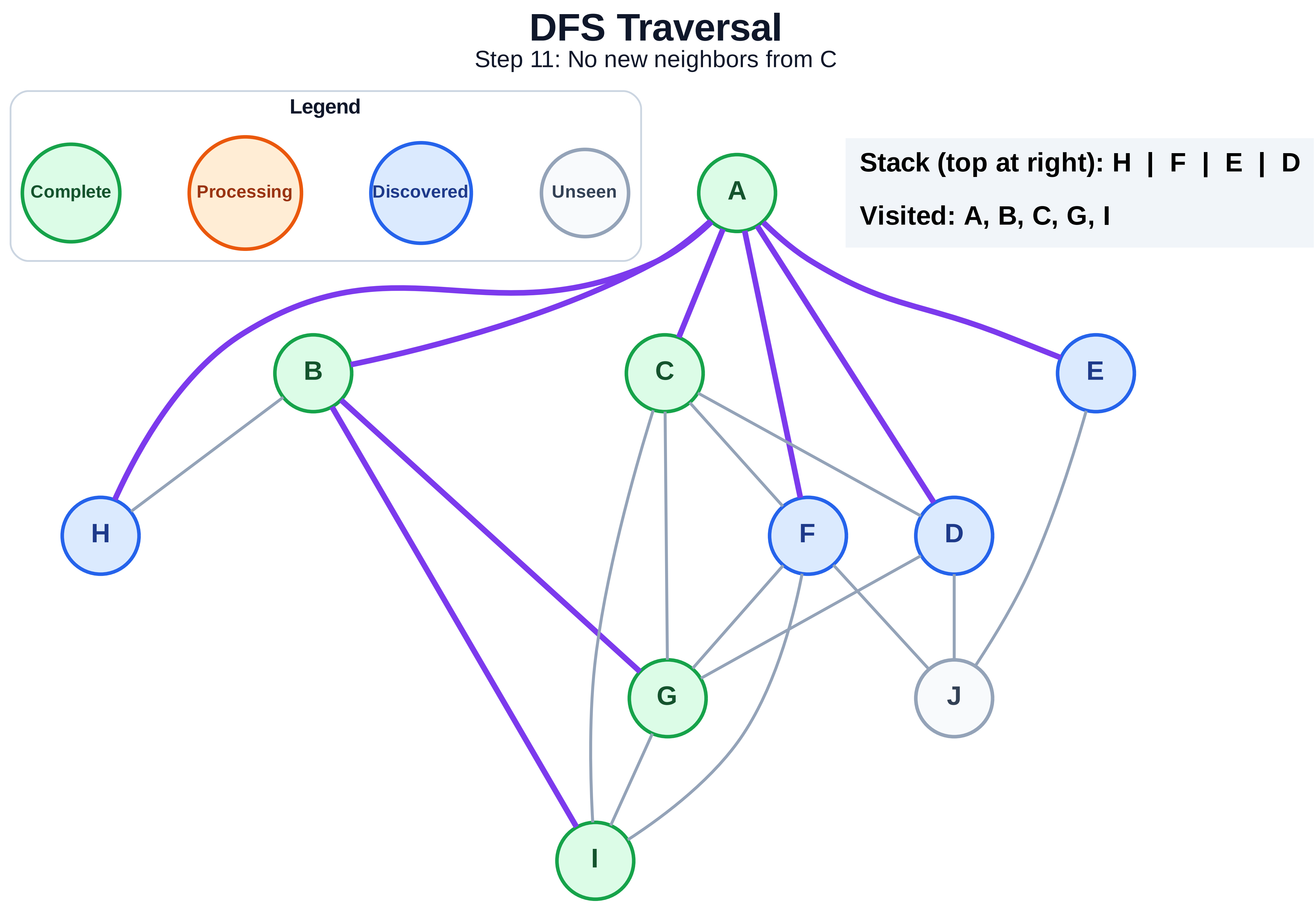
Step 9: No new neighbors from I



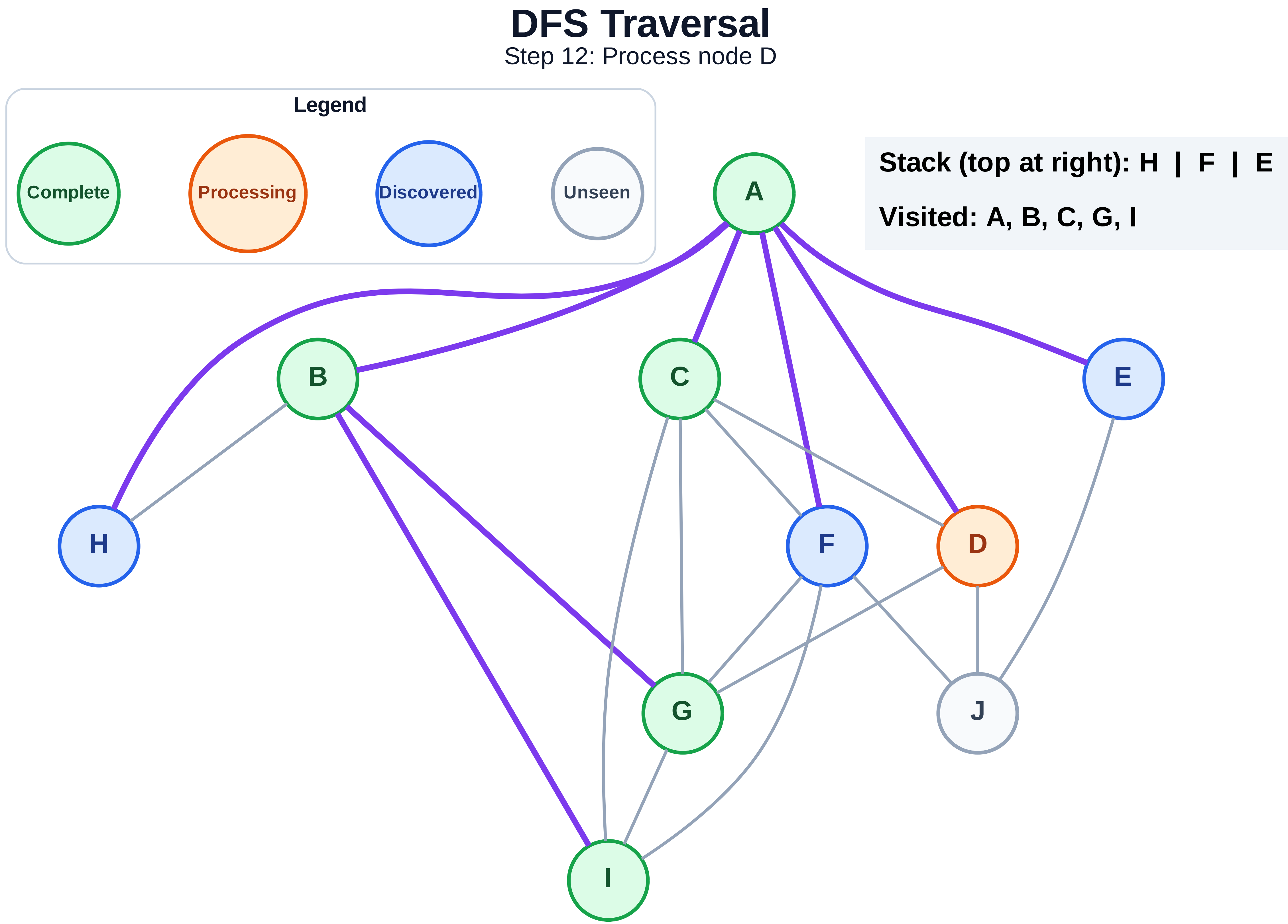
DFS Traversal

Step 10: Process node C



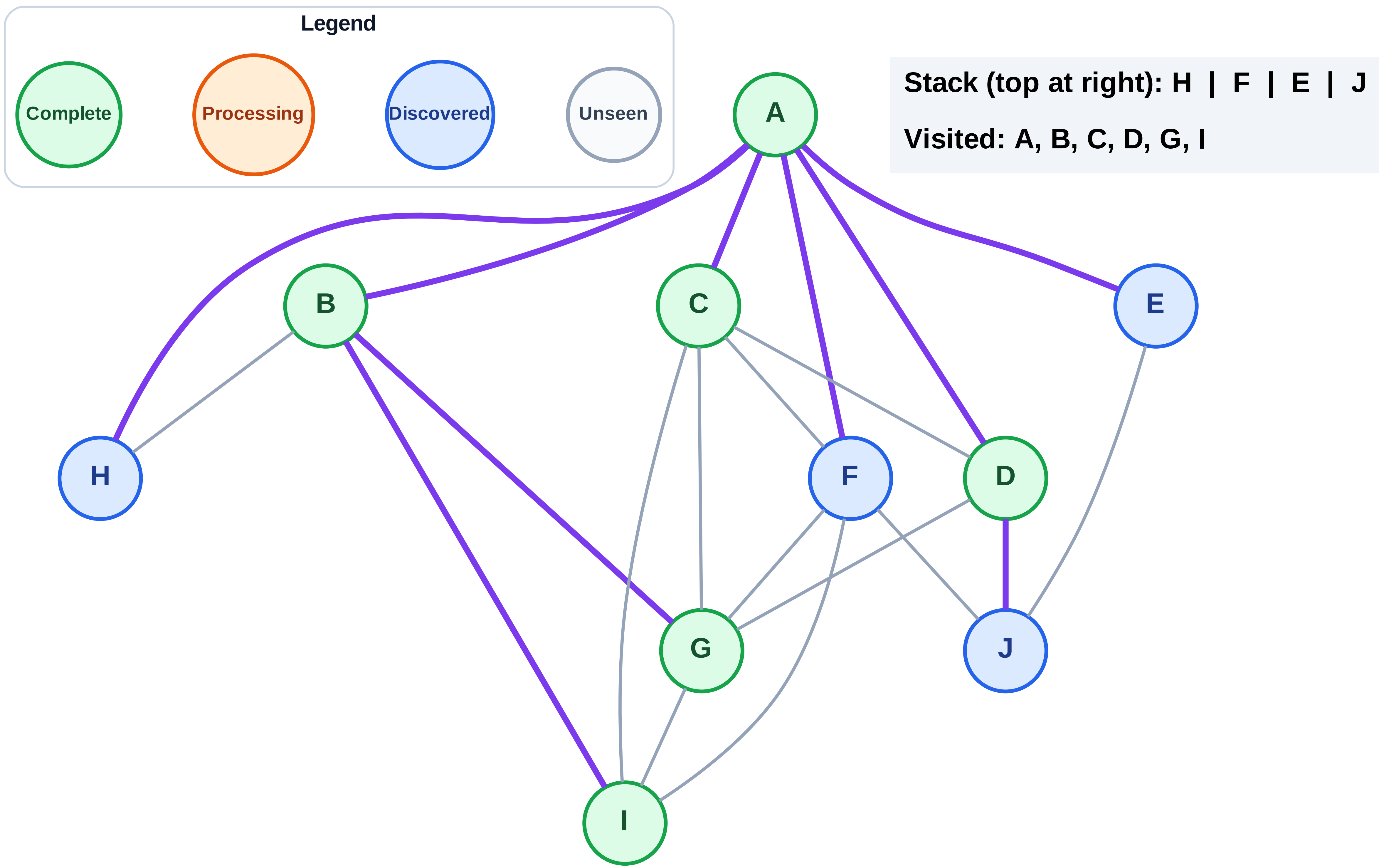


DFS Traversal
Step 11: No new neighbors from C
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): H | F | E | D
Visited: A, B, C, G, I
A
B
C
E
H
F
D
G
J
I


DFS Traversal
Step 12: Process node D
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): H | F | E
Visited: A, B, C, G, I
A
B
C
D
E
F
G
H
I
J

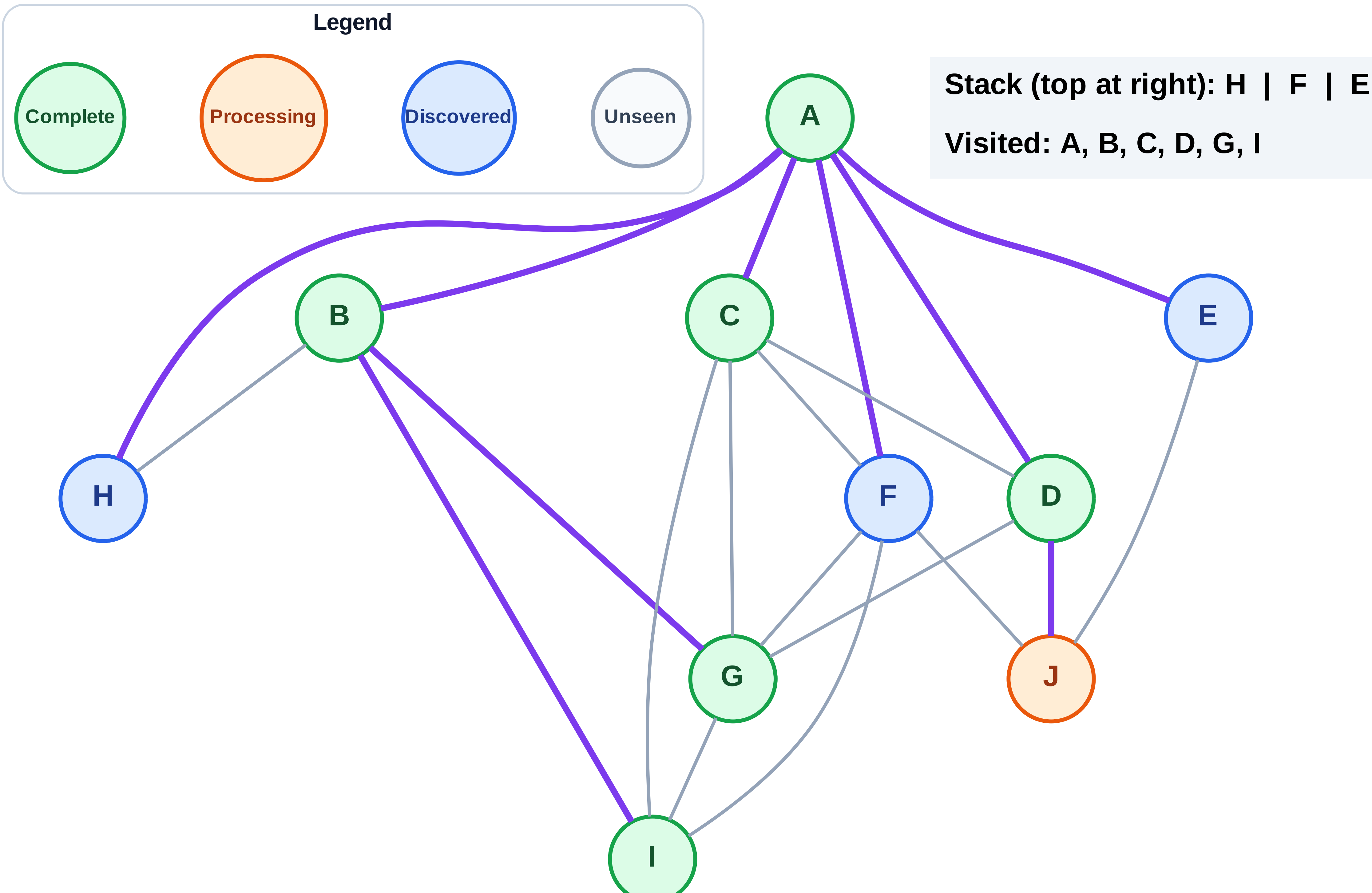
DFS Traversal

Step 13: Discover J from D



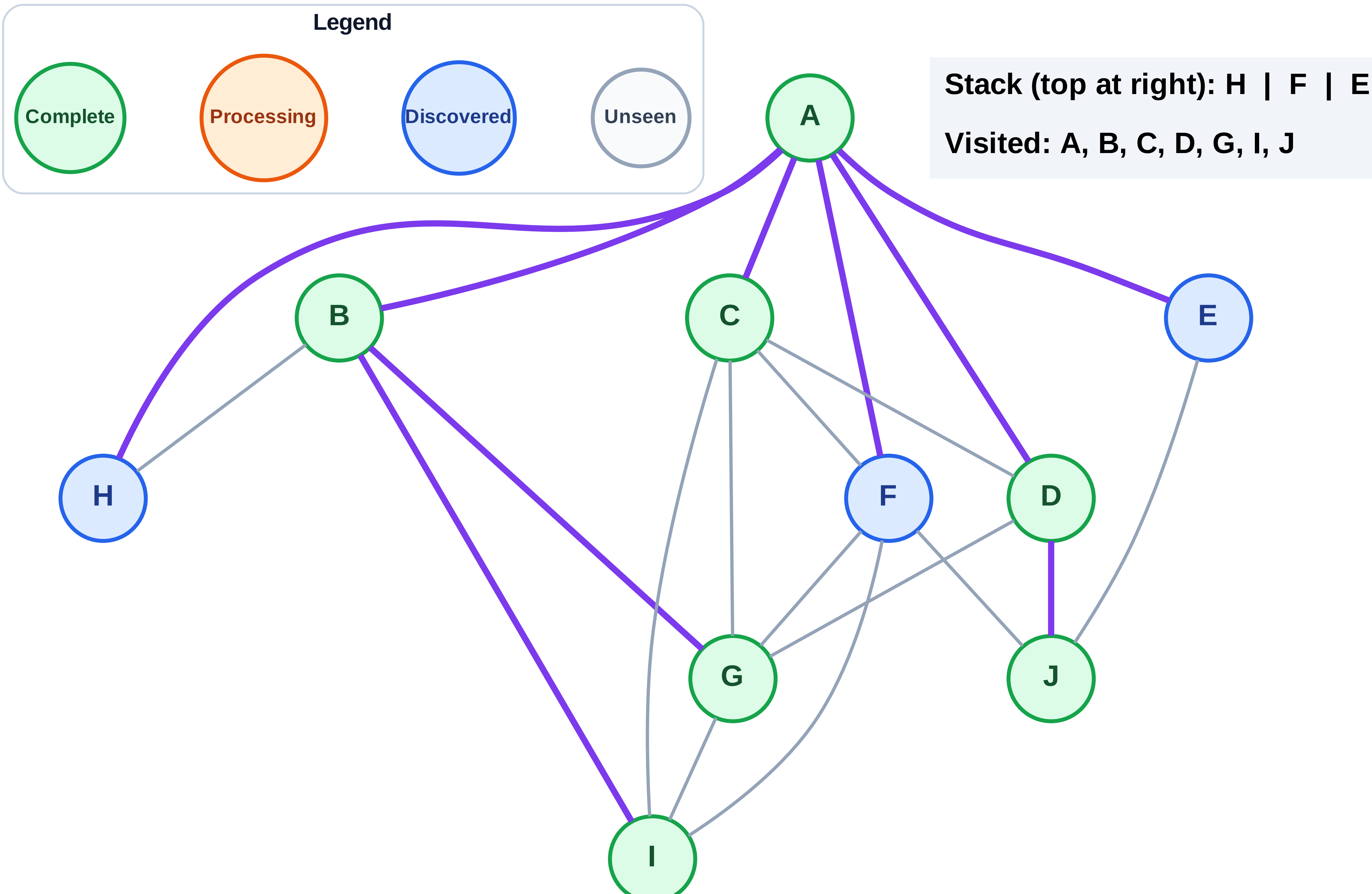
DFS Traversal

Step 14: Process node J



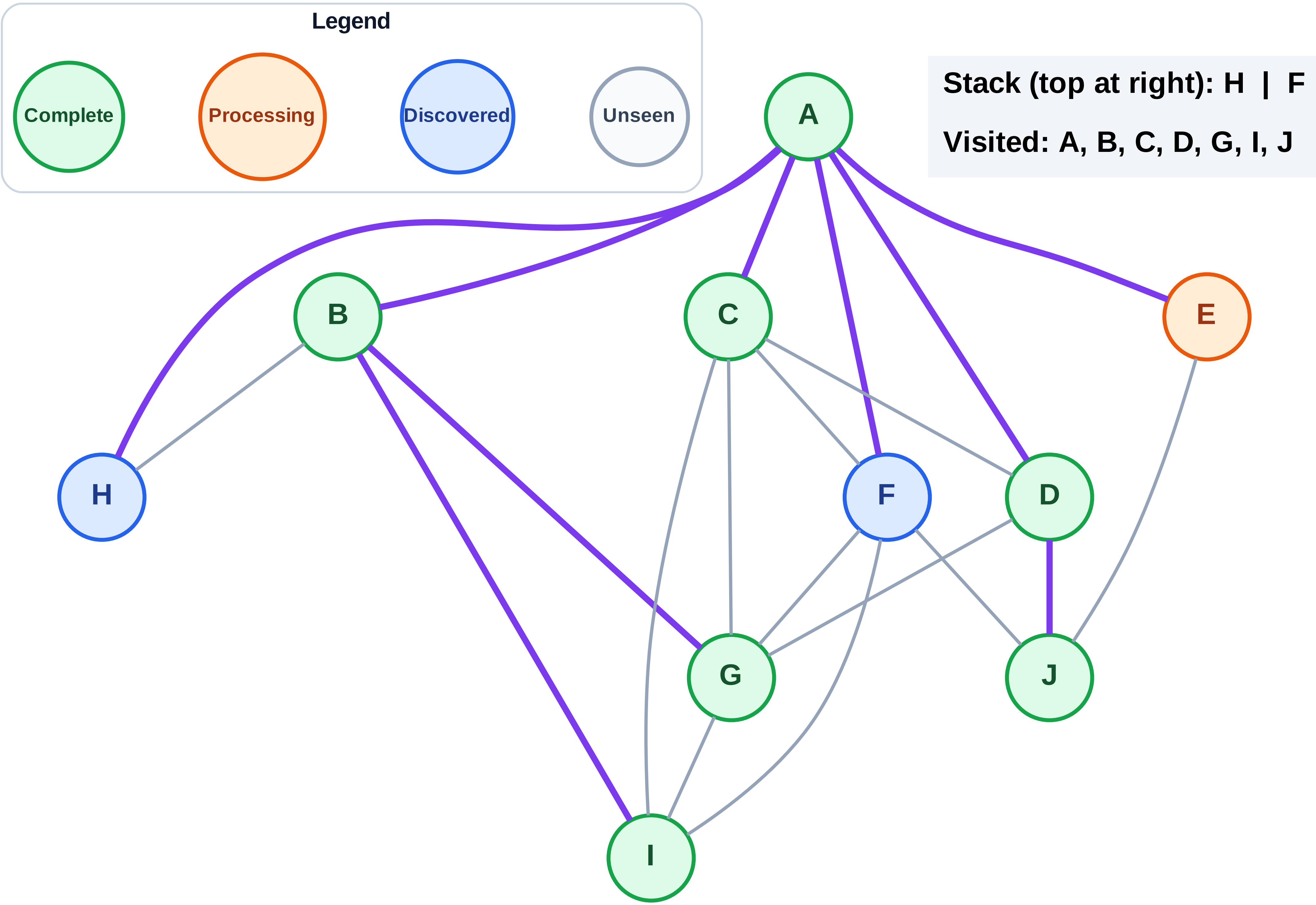
DFS Traversal

Step 15: No new neighbors from J



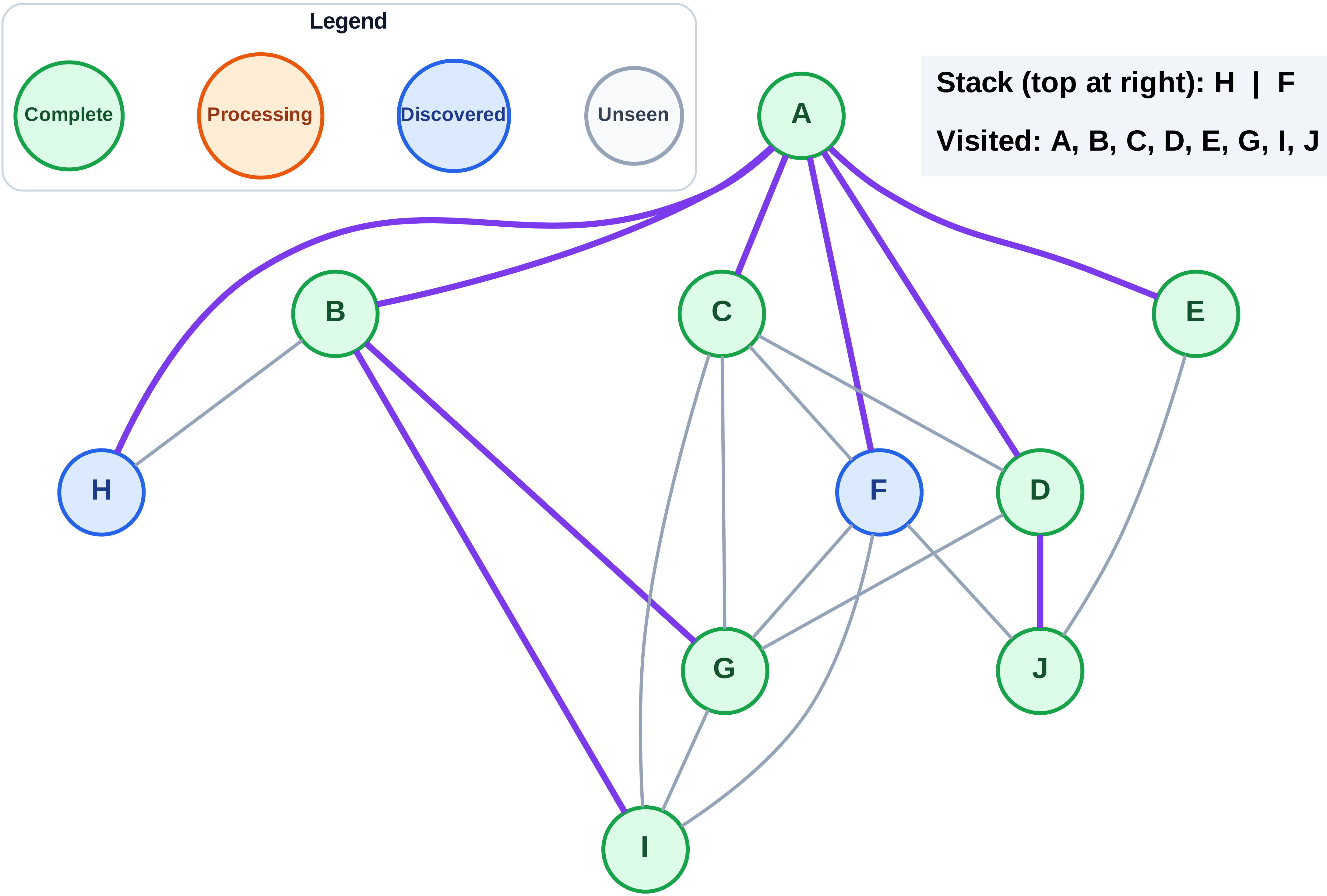
DFS Traversal

Step 16: Process node E



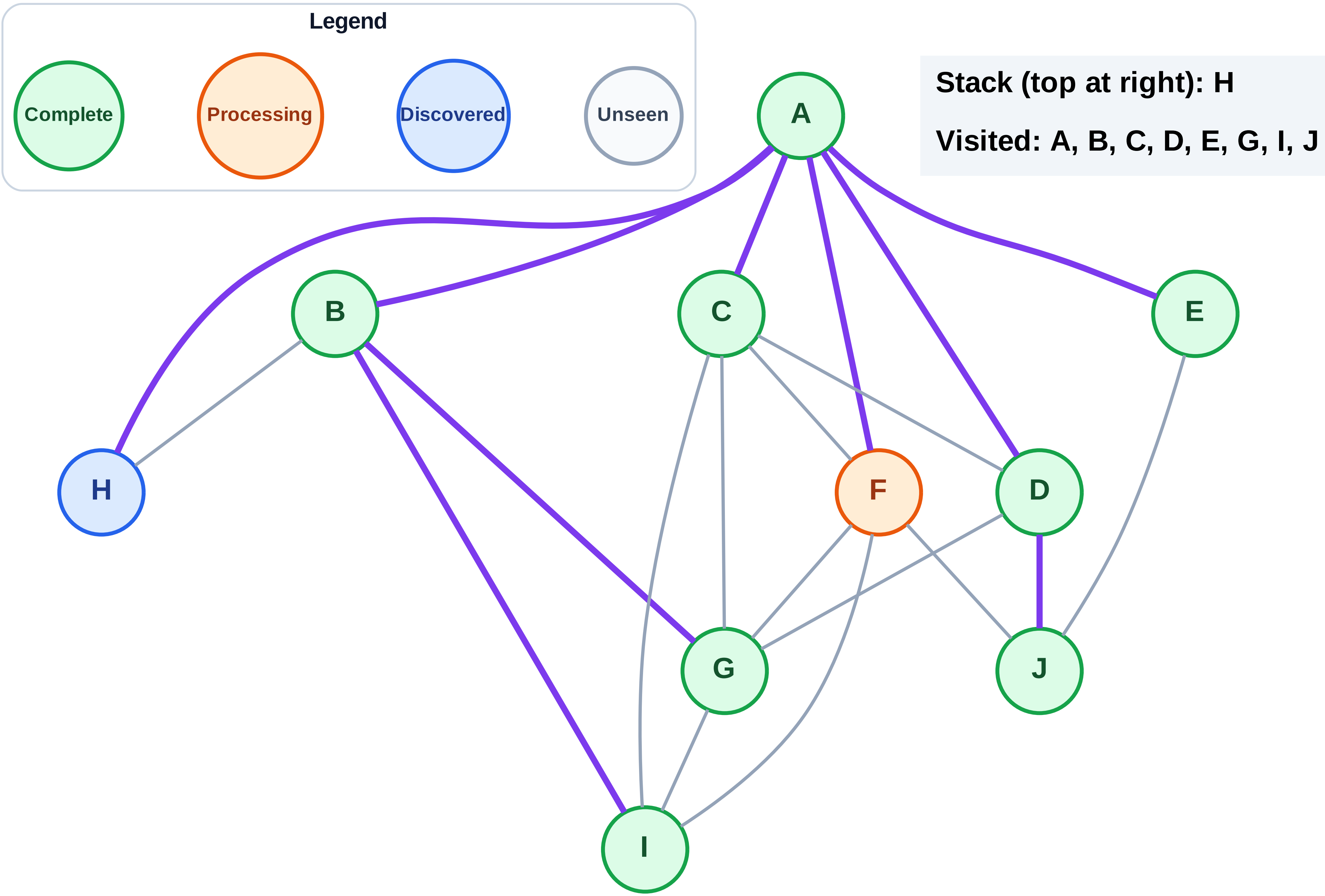
DFS Traversal

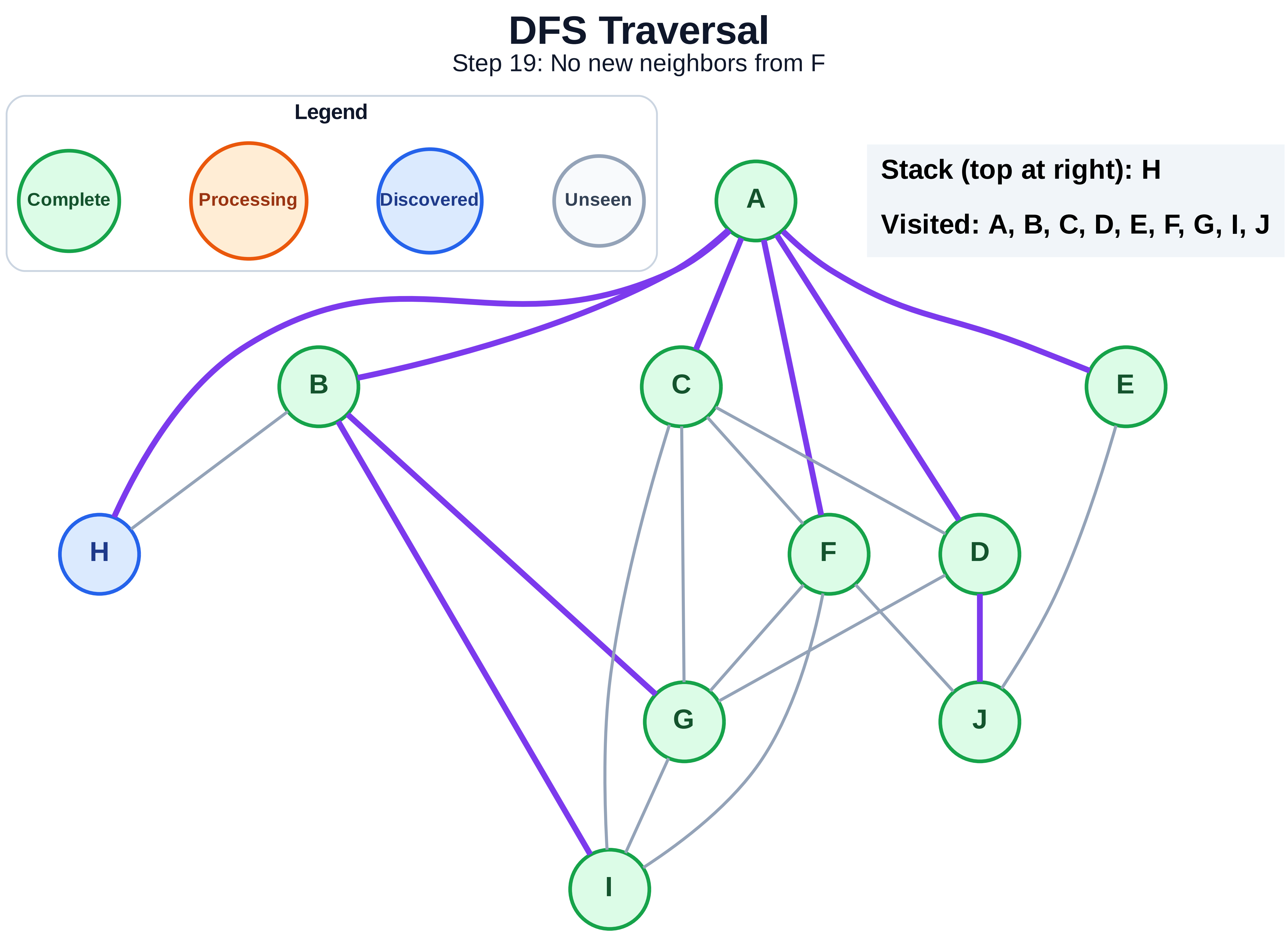
Step 17: No new neighbors from E



DFS Traversal

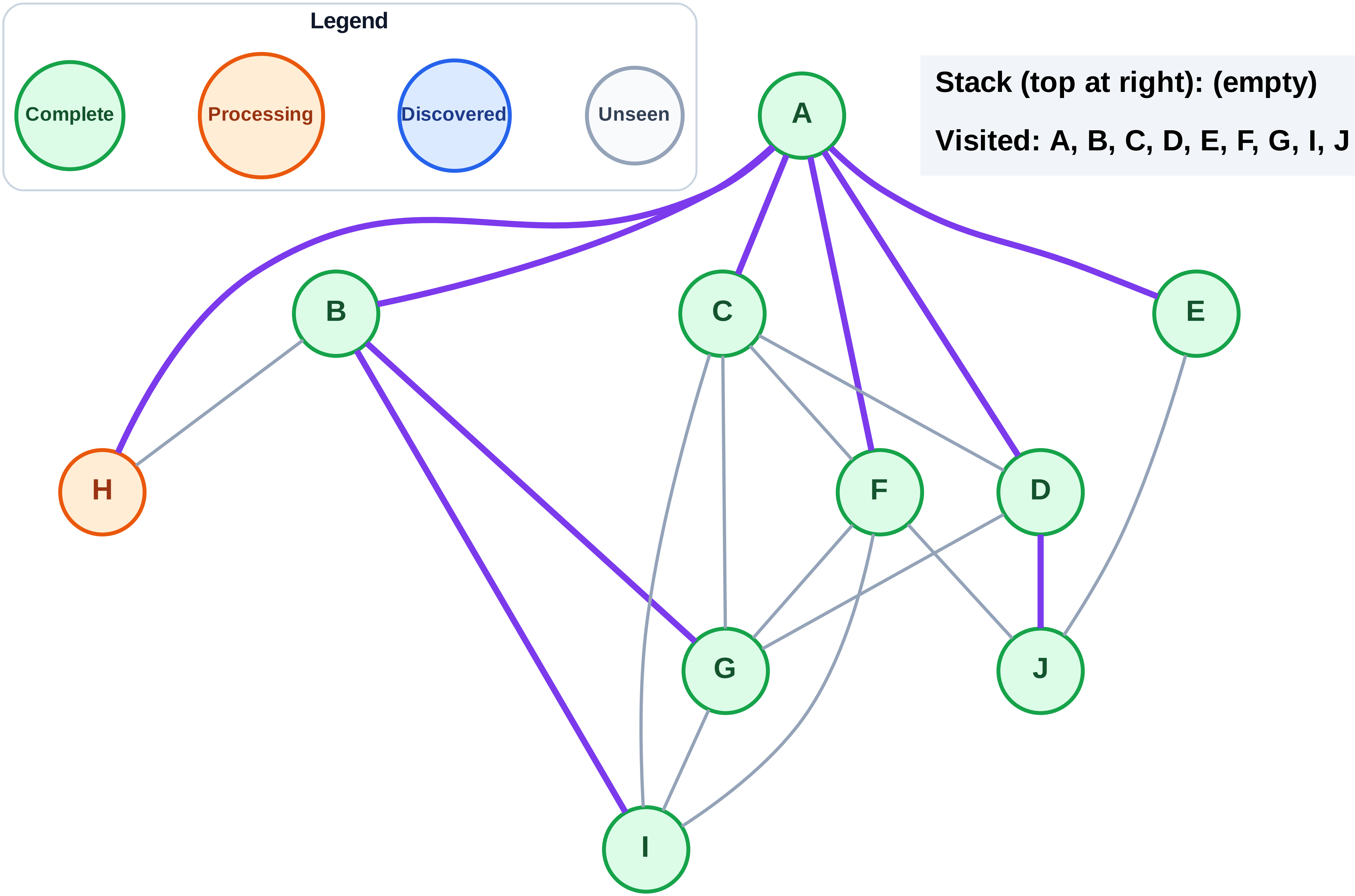
Step 18: Process node F




DFS Traversal
Step 19: No new neighbors from F
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): H
Visited: A, B, C, D, E, F, G, I, J
A
B
C
D
E
F
G
H
I
J

DFS Traversal

Step 20: Process node H



DFS Traversal

Step 21: No new neighbors from H